



PI to PI Interface - APS Connector User Guide

Version 1.5.1.2

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PI to PI Interface - APS Connector User Guide

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Chapter 1. Introduction

This user guide describes the operation of the PI AutoPointSync (APS) Connector for the PI to PI Interface. The informal name of the interface is PItoPI and the informal name of this APS Connector is PItoPI_APS.

To understand this user guide, you must be familiar with the terminology in the [Terminology](#) appendix. Specifically, you must have read the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide that explains important PI AutoPointSync concepts and provides full definitions of PI AutoPointSync terminology.

Note: OSIsoft is revising product documentation and other literature to reflect the evolution of the PI Server from a single server to a multi-server architecture. Specifically, the original historian core of the PI Server is now referred to as the PI Data Archive.

Originally, the PI Server was a single server that contained the PI Data Archive and other subsystems. To add features and improve scalability, the PI Server has evolved from a single server to multiple servers. While the PI Data Archive remains a core server of the PI Server product, the product name “PI Server” now refers to much more than the PI Data Archive. OSIsoft documentation, including this user guide, is changing to use “PI Server” in this broader sense and “PI Data Archive” to refer to the historian core.

The PItoPI_APS Connector is different from all other APS Connectors because the data source is a second PI Data Archive. Therefore, references to the PI Data Archives in this user guide are qualified to distinguish between the two. The *target* PI Data Archive contains the interface points for the PItoPI Interface instance. Each interface point on the target PI Data Archive is configured to collect data from a unique point on the *source* PI Data Archive. The source PI Archive points provide time-series data for the interface points on the target PI Data Archive.

References to PI points in this user guide also are qualified to indicate which PI Data Archive contains each point. The terms *interface point* and *existing point* indicate a point in the target PI Data Archive. The term *source point* indicates a point in the source PI Data Archive.

APS is a tool for synchronizing the PI points for interface instances with the tag definitions in the data sources associated with the interface instances. APS is based on Microsoft’s Component Object Model (COM) technology and includes the following applications and modules:

- **The APS Configuration Utility.**
The APS Configuration Utility is the interactive application that registers instances of interfaces that have corresponding APS Connectors for automatic point synchronization and configures APS options.
- **The APS Synchronization Engine (Sync Engine).**
The APS Synchronization Engine schedules synchronization scans and performs point attribute synchronization for all interface instances that are registered with APS. This module is the foundation of APS and does most of the work. The APS

Synchronization Engine is installed as a Windows service that starts automatically and runs continuously.

- **The APS Synchronization Trigger Service.**
The APS Synchronization Trigger service monitors for events that are configured to trigger a synchronization scan for a registered interface instance. The APS Synchronization Trigger service is installed as a Windows service that starts automatically and runs continuously.
- **APS Connectors (Interface-specific Connectors).**
APS Connectors are interface-specific modules that communicate with the same data source as the interface to obtain current tag attribute values. During each synchronization scan, the APS Synchronization Engine calls the APS Connector for the interface instance to retrieve a list of current tag definitions in the data source. The APS Synchronization Engine uses this information for updating the PI point database.
- **APS Connector-specific Configuration Controls.**
Each APS Connector-specific Configuration Control provides the user interface for configuring the options of a specific APS Connector. If the APS Connector for an interface instance has a connector-specific configuration control, the APS Configuration Utility makes it available when the interface instance is selected for configuration.

The modularized architecture of APS allows a single instance of the APS Configuration Utility and APS Synchronization Engine to support multiple APS Connector modules on one computer. Also, adding new APS Connector modules does not require changes to the APS Configuration Utility or APS Synchronization Engine. The use of COM allows individual components to be upgraded independently of the other components.

When synchronizing the PI points for an interface instance, the APS Synchronization Engine uses an interface-specific APS Connector to obtain current tag definitions from the data source. For the PItoPI Interface, the APS Synchronization Engine calls the PItoPI_APS Connector to retrieve a list of the current tag definitions from the source PI Data Archive. This information is used for creating, updating, or deleting PI points for the PItoPI Interface instance registered with the PItoPI_APS Connector.

APS provides many options to configure its operation for each registered interface instance. Some options apply to the interface instance, such as the interval between synchronization scans or the changes that the APS Synchronization Engine is permitted to make automatically. Other options apply to individual PI points, such as whether the APS Synchronization Engine is allowed to synchronize the point as a whole and, if so, which specific attributes are editable. The APS Synchronization Engine logs all automated actions and changes for auditing by the administrator.

Note: When an interface instance is registered with APS, APS synchronization rules are preset to non-automatic selections and the default per-point synchronization settings are preset to disable synchronization on each point.

The preset options are designed to prevent APS from changing the PI point database until explicitly enabled. While the preset selections for default per-point synchronization settings are in effect, synchronization is disabled for points created or discovered by APS until you manually enable synchronization for those points.

Reference Manuals

OS/soft

- *PI AutoPointSync for Interfaces and PI COM Connectors* (PI APS User Manual.pdf)
- *PI Interface Configuration Utility* (PI Interface Configuration Utility.pdf)
- *PI to PI Interface User Guide* (PI_PItPI.pdf)

Summary of Features and Requirements

The following table summarizes the main features and requirements of the PItoPI_APS Connector. An asterisk (*) in the table indicates that additional explanation follows the table.

Feature	Support
Part Number	PI-IN-OS-PI-NTI
PI Point Types	int16, int32, float16, float32, float64, digital, string
Interface Instance ID Attribute	Location1
Point Class	Classic (see the Point Class section)
Synchronizable PI Attributes	See the table found in the Supported PI Point Attributes section
Must Install on target PI Data Archive Node	No
Must Install on source PI Data Archive Node	No
Must Install on Interface Node	No
* Tag Selection Conditions	No
* Tag Naming Rules	No
Attribute Formulas	No
Attribute Lookup	No
* Additional PI Software Required	Yes
Vendor Software Required on APS Node	No
Vendor Software Required on Foreign Device	No
Vendor Hardware Required	No
Additional PI Software Included with APS Connector	No

Tag Selection Conditions

The PItoPI_APS Connector does not support the standard APS Tag Selection feature. However, the PItoPI_APS Connector supports a configurable WHERE clause that selects a

subset of points on the source PI Data Archive to consider for creation of new interface points. See the [Source Point Filter](#) section for additional information.

Tag Naming Rules

The PItoPI_APS Connector does not support the standard APS Tag Naming feature. However, the PItoPI_APS Connector provides a configurable pattern for constructing tag names for interface points. See the [Tag Naming Convention](#) section for additional information.

Additional PI Software Required

APS obtains information about installed interface instances from configuration data stored in the Module Database by the PI Interface Configuration Utility (ICU). Also, APS stores its configuration information in the Module Database. Therefore, APS requires installation of the OSIsoft products in the following list. In general, these additional OSIsoft products can be installed on separate computers. See the [Diagram of PI APS for the PItoPI Interface](#) section for restrictions that apply to the PItoPI_APS Connector.

- AutoPointSync version 1.2.5.0 or later is required because the PItoPI_APS Connector supports PI security attributes PtSecurity and DataSecurity.
- PI Data Archive version 3.3.361.43 or later is required for Module Database support.
- PI to PI Interface (see the release notes for PItoPI Interface versions that are compatible with this version of PItoPI_APS).
- ICU is required because an interface instance must be managed by ICU before it can be registered with APS. An instance of a Windows interface must be managed by a copy of ICU that is installed on the same computer as the interface instance.

Supported PI Point Attributes

The following table lists the PI point attributes that the PItoPI_APS Connector supports. The PItoPI_APS Connector returns these attributes to the APS Synchronization Engine to use for editing existing points or creating new interface points. An asterisk (*) in the table indicates that additional explanation follows the table.

Attributes that are not in this table are not changed when APS (either the APS Synchronization Engine or APS Configuration Utility) synchronizes an existing point.

When APS creates a new interface point, default values are assigned to attributes that are not in the table. APS has configurable default values for some attributes that have no corresponding attributes in most data sources (see the [Security & Archive Settings](#) section). Other than these special attributes, the default values configured in the target PI Data Archive are assigned to attributes that are not in the table.

Note: When a new interface point is created, attributes not in the table are set to default values by either APS (for the special cases) or the target PI Data Archive (for all others).

Source PI Data Archive Attribute Column

The *Source PI Data Archive Attribute* column contains the data source attribute that provides the value for each PI attribute.

The notation <blank> indicates that the value is a zero-length string.

The notation <APS Config> in this column indicates that the PItoPI_APS Connector provides an option to ignore the value for this attribute from the source point (see the [Point Attributes](#) section). When the option is selected, the PItoPI_APS Connector does not include the attribute in the attribute values that are returned to the APS Synchronization Engine for a point. As a result, if the source point corresponds to an existing interface point, APS does not change the interface point attribute. However, if APS creates a new interface point for the source point, APS provides a configurable default value for the attribute. The configurable default value can be viewed and changed with the APS Configuration Utility (on the **Security & Archive Settings** tab in the **User-set Defaults** dialog box).

Sync Column

The *Sync.* column indicates whether an attribute is used for point creation only or for both point creation and synchronization of existing points. Notice that APS always uses the attribute value from the APS Connector for new interface point creation.

Note: The per-point synchronization settings do *not* apply to new interface point creation.

No in this column indicates that the attribute cannot be synchronized. That is, the APS Connector provides a value for this attribute for new interface point creation but does not provide a value for editing an existing point. Generally, the key attributes for the APS Connector (and interface) are not synchronizable because changing a key attribute causes an existing point to reference a different source point. The PItoPI_APS Connector is an exception because any one of the key attributes is sufficient to identify the source point. Therefore, the key attributes that do not identify the source point are synchronizable (with a few constraints that are discussed later in this document).

“Yes” in this column indicates that the APS Connector can provide a value for this attribute for both new interface point creation and editing the attribute of an existing point. The attribute value from the APS Connector is always used for new interface point creation. However, the per-point synchronization settings control whether APS can edit the attribute for a specific existing point.

When the PItoPI_APS Connector retrieves attributes for a source point in a point class that does *not* include the Classic attribute set, the handling of the Classic attributes depends on whether the PItoPI_APS Connector was called to find available points for creation or to get current source point attributes for synchronizing existing interface points.

- If the PItoPI_APS Connector was called for available points, the missing Classic attributes are set to the built-in values in the “Default Value” column.
- If the PItoPI_APS Connector was called to get current source point attributes for existing target points, no value is returned for the missing Classic attributes, which the APS Synchronization Engine treats as unchanged attributes. That is, the missing Classic attributes are not synchronized.

Default Value Column

The *Default Value* column in the table shows the default values that the APS Connector provides if the source point does not provide a value.

For the PItoPI_APS Connector, points in the source PI Data Archive are usually members of a point class that contains the Classic attribute set and the PItoPI_APS Connector is able to return the value in the *Source PI Data Archive Attribute* column for every supported attribute. The *Default Value* column only applies when a point in the source PI Data Archive belongs to a point class that does not contain the Classic attribute set (see the [Limited Support of Custom Point Classes](#) section). In this case, the PItoPI_APS Connector only requests the attributes in the table that are in the Base point class (indicated by *n/a* in the *Default Value* column).

Description	PI Point Attribute	Source PI Data Archive Attribute	Sync.	Default Value
PI point name	Tag	Tag See Tag Naming Conventions section	Yes	n/a
PI point type	PointType	PointType See Point Type section	No	n/a
Source PI Data Archive Tag Name	* InstrumentTag	Tag, <blank>, or InstrumentTag (see note)	Yes	
Extended point description	* ExDesc	Tag or ExDesc (see note)	Yes	n/a
Source PI Data Archive PointID	* UserInt1	PointID, 0, or UserInt1 (see note)	Yes	0
Interface ID	* Location1	Interface ID or Location1	Yes	0
Data timestamp adjustment	* Location2	0 or Location2	Yes	0

Description	PI Point Attribute	Source PI Data Archive Attribute	Sync.	Default Value
Digital state suppression	* Location3	3 or Location3	Yes	0
Scan class	* Location4	Scan class or Location4	Yes	0
Edit handling option	* Location5	0 or Location5	Yes	0
Point description	Descriptor	Descriptor	Yes	n/a
Engineering units	EngUnits	EngUnits	Yes	n/a
Point minimum value	Zero	Zero	Yes	n/a
Point value range	Span	Span	Yes	n/a
Digital state set name	* DigitalSet	DigitalSet	Yes	n/a
States in digital state set	* DigitalStateSet		Yes	n/a
Source tag for output points	SourceTag	SourceTag or <blank>	Yes	n/a
Stepped changes	Step	Step	Yes	n/a
Collect point data	Scan	Scan	Yes	n/a
Shutdown Events	Shutdown	Shutdown	Yes	n/a
Archiving flag	Archiving	Archiving	Yes	n/a
Compress point data	Compressing	Compressing or <APS Config>	Yes	n/a
Compression minimum time between events	CompMin	CompMin or <APS Config>	Yes	n/a
Compression maximum time between events	CompMax	CompMax or <APS Config>	Yes	n/a
Compression deviation in engineering units	* CompDev	CompDev or <APS Config>	Yes	n/a
Compression deviation as a percent of span	* CompDevPercent	CompDevPercent or <APS Config>	Yes	n/a
Exception minimum time between events	ExcMin	ExcMin or <APS Config>	Yes	n/a
Exception maximum time between events	ExcMax	ExcMax or <APS Config>	Yes	n/a
Exception deviation in engineering units	* ExcDev	ExcDev or <APS Config>	Yes	n/a
Exception deviation as a percent of span	* ExcDevPercent	ExcDevPercent or <APS Config>	Yes	n/a
Data access owner	DataOwner	DataOwner or <APS Config>	Yes	n/a
Data access group	DataGroup	DataGroup or <APS Config>	Yes	n/a
Data access permissions	DataAccess	DataAccess or <APS Config>	Yes	n/a
Data access security	DataSecurity	DataSecurity or <APS Config>	Yes	n/a

Description	PI Point Attribute	Source PI Data Archive Attribute	Sync.	Default Value
Point access owner	PtOwner	PtOwner or <APS Config>	Yes	n/a
Point access group	PtGroup	PtGroup or <APS Config>	Yes	n/a
Point access permissions	PtAccess	PtAccess or <APS Config>	Yes	n/a
Point access security	PtSecurity	PtSecurity or <APS Config>	Yes	n/a
Typical value for point	TypicalValue	TypicalValue	Yes	n/a
Display digits	DisplayDigits	DisplayDigits	Yes	n/a
Interface-specific	TotalCode	TotalCode	Yes	0
Interface-specific	SquareRoot	SquareRoot	Yes	0
Interface-specific	Convers	Convers	Yes	1
Filter code	* FilterCode	FilterCode or 0	Yes	0
Interface-specific	UserInt2	UserInt2	Yes	0
Interface-specific	UserReal1	UserReal1	Yes	0.0
Interface-specific	UserReal2	UserReal2	Yes	0.0
Future tag	Future	Future	Yes	n/a

InstrumentTag

Depending on the PItPI Interface configuration, the interface either uses the InstrumentTag attribute as a key attribute or ignores the InstrumentTag attribute. (See the [How PItPI and PItPI APS Find Source Points](#) section for the complete explanation.) The PItPI_APS Connector must provide a value for the InstrumentTag attribute that is consistent with how the interface uses the attribute. Therefore, the PItPI_APS Connector analyzes the interface parameters to determine the appropriate value to return to the APS Synchronization Engine for the InstrumentTag attribute.

The following summary does not include all of the special cases that the PItPI_APS Connector considers in determining the new value for the InstrumentTag attribute. The details are presented with the configuration option for the InstrumentTag attribute under [InstrumentTag](#) in the *Connector-specific Configuration Control* chapter.

If the interface uses InstrumentTag as a key attribute, the InstrumentTag value must contain the source point tag name or an empty string. The PItPI_APS Connector can be configured to provide the source point tag name or an empty string for the InstrumentTag attribute value.

If the interface ignores the InstrumentTag attribute, the interface point InstrumentTag value can be synchronized with the source point InstrumentTag value. The PItPI_APS Connector can be configured to provide the source point tag name or InstrumentTag value for the InstrumentTag attribute value.

ExDesc

Depending on the PItPI Interface configuration, the interface either uses the ExDesc attribute as a key attribute or ignores the ExDesc attribute. (See the [How PItPI and](#)

[PItoPI APS Find Source Points](#) section for the complete explanation.) The PItoPI_APS Connector must provide a value for the ExDesc attribute that is consistent with how the interface uses the attribute. Therefore, the PItoPI_APS Connector analyzes the interface parameters to determine the appropriate value to return to the APS Synchronization Engine for the ExDesc attribute.

The following summary does not include all of the special cases that the PItoPI_APS Connector considers in determining the new value for the ExDesc attribute. The details are presented with the configuration option for the ExDesc attribute under [ExDesc](#) in the *Connector-specific Configuration Control* chapter.

If the interface uses ExDesc as a key attribute, the interface checks for the source point tag name in the ExDesc value. If the ExDesc value begins with a source point tag name specification of the form *STAG=tag*, *tag* is the source point tag name. Otherwise, the ExDesc attribute is ignored. The PItoPI_APS Connector can be configured to provide either a source point tag name specification or the source point ExDesc value for the ExDesc value. Before providing the source point ExDesc value, however, the PItoPI_APS Connector determines if the interface may find a source point tag name specification in the value. If so, the PItoPI_APS Connector modifies the source point ExDesc value so that the interface will not recognize the source point tag name specification.

If the interface ignores the ExDesc attribute, the interface point ExDesc value can be synchronized with the source point ExDesc value. The PItoPI_APS Connector can be configured to provide a source point tag name specification or the source point ExDesc value for the ExDesc attribute value.

UserInt1

Depending on the PItoPI Interface configuration, the interface either uses the UserInt1 attribute as a key attribute or ignores the UserInt1 attribute. (See the [How PItoPI and PItoPI APS Find Source Points](#) section for the complete explanation.) The PItoPI_APS Connector must provide a value for the UserInt1 attribute that is consistent with how the interface uses the attribute. Therefore, the PItoPI_APS Connector analyzes the interface parameters to determine the appropriate value to return to the APS Synchronization Engine for the UserInt1 attribute.

The following summary does not include all of the special cases that the PItoPI_APS Connector considers in determining the new value for the UserInt1 attribute. The details are presented with the configuration option for the UserInt1 attribute under [UserInt1](#) in the *Connector-specific Configuration Control* chapter.

If the interface uses UserInt1 as a key attribute, the UserInt1 value must contain the source point ID or a number less than one (which is not a PI point ID because all PI point IDs are greater than zero). The PItoPI_APS Connector can be configured to provide the source point ID or UserInt1 value for the UserInt1 attribute value. Before providing the source point UserInt1 value, however, the PItoPI_APS Connector determines if the UserInt1 value is greater than zero, which the interface may interpret as a source point ID. If so, the PItoPI_APS Connector returns zero (instead of the source point UserInt1 value) so that the interface will not interpret UserInt1 as the source point ID.

If the interface ignores the UserInt1 attribute, the interface point UserInt1 value can be synchronized with the source point UserInt1 value. The PItoPI_APS Connector can be configured to provide the source point ID or UserInt1 value for the UserInt1 attribute value.

Recommendation: Configure the PItoPI_APS Connector to store the source point ID in the UserInt1 attribute. This option is required for the PItoPI_APS Connector to reconfigure an interface point when the corresponding source PI Data Archive point is renamed.

If a point on the source PI Data Archive is renamed, an interface point that references the original source point tag name will fail to load when the interface point is edited or the PItoPI Interface starts. For the PItoPI_APS Connector to reconfigure an interface point when the corresponding source PI Data Archive point is renamed, the source point ID must be stored in the interface point, specifically in the UserInt1 attribute.

Note: For APS to store the source point ID in the UserInt1 attribute of existing interface points, the UserInt1 attribute must be enabled for synchronization for all interface points.

The [PItoPI_APS Connector Extensions](#) section explains how the PItoPI_APS Connector distinguishes between a deleted or renamed source point when an interface point references a nonexistent source point tag name.

Location1

Location1 is the interface ID attribute for the PItoPI Interface. Since the Location1 setting for the source PI Data Archive point is usually an interface-specific value for some other interface, the Location1 attribute is normally set to the interface ID from the interface configuration by APS when a new interface point is created and not changed when an existing point is synchronized. However, if the interface command line includes the `/c1` (override Location1) parameter, then the Location1 attribute can be set from or synchronized with the Location1 attribute from the source point.

Location2

Location2 configures how the PItoPI Interface handles timestamps. Since the Location2 setting for the source PI Data Archive point is usually an interface-specific value for some other interface, the Location2 attribute is normally set to 0 for new interface points and not changed when an existing point is synchronized. However, if the interface command line includes the `/c2=x` (override Location2) parameter, then the Location2 attribute can be set from or synchronized with the Location2 attribute from the source point.

Location3

Location3 configures how the PItoPI Interface handles digital states. Since the Location3 setting for the source PI Data Archive point is usually an interface-specific value for some other interface, the Location3 attribute is normally set to 3 for new interface points and not changed when an existing point is synchronized. However, if the interface command line includes the `/c3=x` (override Location3) parameter, then the Location3 attribute can be set from or synchronized with the Location3 attribute from the source point.

Location4

Location4 is the scan class. Since the Location4 setting for the source PI Data Archive point is usually an interface-specific value for some other interface, the Location4 attribute is normally set to a configurable scan class (default is scan class 1) and not changed when an existing point is synchronized. However, if the interface command line includes the `/c4=x` (override Location4) parameter, then the Location4 attribute can be set from or synchronized with the Location4 attribute from the source point.

Location5

Location5 configures how the PItoPI Interface handles manual edits of point data. Since the Location5 setting for the source PI Data Archive point is usually an interface-specific value for some other interface, the Location5 attribute is normally set to 0 for new interface points and not changed when an existing point is synchronized. However, if the interface command line includes the `/c5=x` (override Location5) parameter, then the Location5 attribute can be set from or synchronized with the Location5 attribute from the source point.

DigitalSet

The DigitalSet attribute only applies to digital points. If the PointType of a source point is not digital, the PItoPI_APS Connector does not return a value for the DigitalSet attribute. For digital source points, the PItoPI_APS Connector provides configurable options for adding a prefix, suffix, or both to digital set names from the source PI Data Archive to construct the DigitalSet attribute value for an interface point. When a PItoPI Interface instance is registered with APS, these options are initially disabled and the PItoPI_APS Connector returns the DigitalSet attribute from the source point as the DigitalSet attribute value for the interface point.

When the PItoPI_APS Connector returns attributes for a digital point, the APS Synchronization Engine checks whether a digital state set already exists in the target PI Data Archive with the name in the DigitalSet attribute.

If the digital state set name does not exist and APS is creating or editing an interface point, APS first creates a digital state set with this name and the list of states in the DigitalStateSet attribute.

If the digital state set name *does* exist, the APS Synchronization Engine compares the state list for the digital state set in the target PI Data Archive with the state list in the DigitalStateSet attribute value from the PItoPI_APS Connector. If the state lists agree, the APS Synchronization Engine accepts the DigitalSet attribute value from the PItoPI_APS Connector (that is, uses the existing digital set). However, if the state list for a digital state set in the target PI Data Archive *disagrees* with the state list in the DigitalStateSet attribute value, the behavior depends on the version of APS. Consult the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide for the specific APS version for an explanation of how disagreement between state lists is handled.

DigitalStateSet

DigitalStateSet is an artificial attribute that APS uses internally; it is not an actual PI attribute. DigitalStateSet only applies to digital points and is coupled with the DigitalSet attribute. For a digital point, the value in the DigitalSet attribute contains the name of the PI digital state set for the point. If the DigitalStateSet attribute for the point is not empty, it contains a list of the states expected in the digital state set.

The PItoPI_APS Connector obtains the digital states from the source PI Data Archive to set the value for the DigitalStateSet artificial attribute.

CompDev and CompDevPercent

The CompDev and CompDevPercent attributes are interrelated in the PI Data Archive; setting or editing one affects the value of the other. When APS creates a new interface point, both attribute values are sent to the target PI Data Archive, which gives precedence to the CompDevPercent attribute.

When APS synchronizes an existing point, the per-point synchronization settings control which attributes are edited. Since the CompDevPercent attribute is expressed as a percent of span, the Span attribute must be enabled for synchronization when the CompDevPercent attribute is enabled.

If settings for an existing point enable synchronization for one of the CompDev and CompDevPercent attributes, the enabled attribute is edited.

If settings for an existing point enable synchronization of both CompDev and CompDevPercent attributes, the target PI Data Archive gives precedence to the CompDevPercent attribute.

Recommendation: Since CompDev is actually stored in the PI Data Archive and CompDevPercent is a derived attribute, disable synchronization of CompDevPercent for all interface points.

ExcDev and ExcDevPercent

The ExcDev and ExcDevPercent attributes are interrelated in the PI Data Archive; setting or editing one affects the value of the other. When APS creates a new interface point, both attribute values are sent to the target PI Data Archive, which gives precedence to the ExcDevPercent attribute.

When APS synchronizes an existing point, the per-point synchronization settings control which attributes are edited. Since the ExcDevPercent attribute is expressed as a percent of span, the Span attribute must be enabled for synchronization when the ExcDevPercent attribute is enabled.

If settings for an existing point enable synchronization for one of the ExcDev and ExcDevPercent attributes, the enabled attribute is edited.

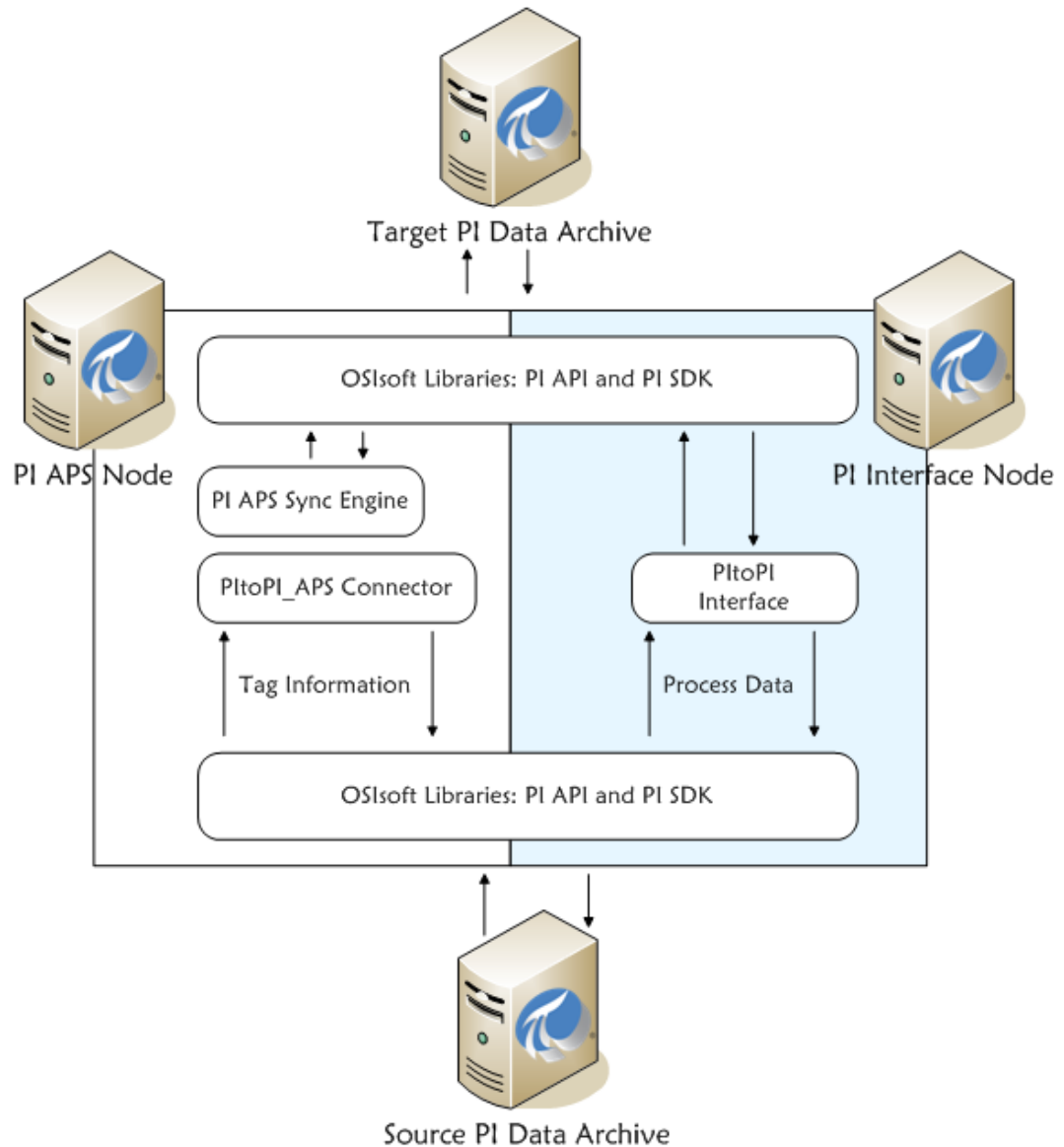
If settings for an existing point enable synchronization of both ExcDev and ExcDevPercent attributes, the target PI Data Archive gives precedence to the ExcDevPercent attribute.

Recommendation: Since ExcDev is actually stored in the PI Data Archive and ExcDevPercent is a derived attribute, disable synchronization of ExcDevPercent for all interface points.

FilterCode

PI3 digital points can only have a FilterCode attribute value of 0. The PItoPI_APS Connector does not pass through the FilterCode attribute from a digital source point but does provide 0 for the FilterCode attribute value for the interface point.

Diagram of PI APS for the PltoPI Interface



The implementation of APS requires installation of the APS Synchronization Engine, APS Configuration Utility, and the APS Connectors on the same computer (identified as *PI APS Node* in the diagram above). ICU must be installed on the PI Interface node. In general, the APS components, target PI Data Archive, interface, and source PI Data Archive can be distributed on four separate computers as depicted in the diagram. If the PI Interface node has low CPU usage, the APS components can be installed on the same node as the PI Interface. OSIsoft generally recommends against installing APS on a PI Data Archive node.

PItoPI_APS Features and Limitations

The following features and limitations are specific to the PItoPI_APS Connector.

Filtering Source Points

The default configuration of the PItoPI_APS Connector considers all points on the source PI Data Archive as candidates for interface points. For many reasons, a PItoPI Interface instance may need to be restricted to a subset of the source PI Data Archive points. The PItoPI_APS Connector provides a configurable WHERE clause to select the source PI Data Archive points to consider for candidate interface points. The PI SDK provides two methods for selectively retrieving points from a PI3 Data Archive: GetPoints and GetPointsSQL. The PItoPI_APS Connector can be configured to use GetPoints or GetPointsSQL. In general, GetPoints is faster but only handles simple WHERE clauses. GetPointsSQL handles complex WHERE clauses but is usually slower. The configuration options are described in the [Source Point Filter](#) section.

Synchronization of Renamed Source Points

When the PItoPI Interface loads a point, the interface must identify the source point from which data will be collected. In the *PI to PI Interface User Guide* and other parts of this user guide, this process is referred to as *mapping* the interface point to a source point.

Typically, PItoPI Interface points are mapped to source PI Data Archive points by source point tag name. A problem associated with using source point tag name to configure an interface point is that renaming the source point causes the interface point to fail when the interface restarts or the interface point is edited. The PItoPI_APS Connector supports synchronizing the interface point attributes that contain source point tag name when the source point is renamed. To make this possible, the PItoPI_APS Connector provides an option to return the source point ID as the UserInt1 attribute value for interface points.

Note: For the PItoPI_APS Connector to synchronize with a renamed source point, the option to return source point ID as the UserInt1 attribute value must be selected and the UserInt1 attribute must be enabled for synchronization for all existing points.

Unless the interface is configured for source PI Data Archive-level failover, using the UserInt1 attribute to hold the source point ID is compatible with the PItoPI Interface. When the interface is configured for source PI Data Archive-level failover, see the [Source PI Data Archive-level Failover Restrictions](#) section for a discussion of interface point configurations that the interface rejects if the UserInt1 attribute contains a value greater than zero, specifically a source point ID.

To obtain current source PI Data Archive point attributes for an existing interface point, the PItoPI_APS Connector first follows the same logic as the PItoPI Interface for determining the source point. If an interface point is configured with source point tag name in an active key attribute and the PItoPI_APS Connector discovers that a source point with the tag name does not exist, the PItoPI_APS Connector checks 1) if the option to store source point ID in UserInt1 is selected, and 2) if the UserInt1 attribute value of the interface point contains the point ID of an existing point on the source PI Data Archive. If both conditions are true, the PItoPI_APS Connector uses the current source point tag name to construct values for the interface point attributes that contain source point tag name. If APS is allowed to synchronize those attributes, the interface point is edited to reference the new source point tag name.

If the option to store source point ID in UserInt1 is not selected or the UserInt1 attribute does not contain an existing source point ID, a renamed source point appears to the PItoPI_APS Connector as if the source point with the original name is deleted and a new point is created with the new name. Therefore, the PItoPI_APS Connector indicates that the existing interface point is deleted and returns an available point for the renamed source point, which causes the APS Synchronization Engine to apply the rule for deleted points to the interface point and apply the rule for creating points to the renamed source point. See the [PItoPI_APS Connector Extensions](#) section for more information.

Limitations

PItoPI_APS Uses the UserInt1 Attribute

To synchronize an interface point if the mapped source PI Data Archive point is renamed, the PItoPI_APS Connector needs the source point ID. Therefore, the PItoPI_APS Connector provides an option to return the source point ID to the APS Synchronization Engine as the current value for the UserInt1 attribute of interface points. If the option to store source point ID in UserInt1 is selected and synchronization of the UserInt1 attribute is enabled, APS changes the UserInt1 attribute to the source point ID. However, the PItoPI_APS Connector does not know whether synchronization of the UserInt1 attribute is enabled for any specific interface point and, therefore, does not know whether the UserInt1 attribute is actually changed to the source point ID. During later synchronization scans, if the PItoPI_APS Connector cannot find a source point by tag name and the option to store source point ID in UserInt1 is selected, the PItoPI_APS Connector assumes that a UserInt1 attribute value greater than zero is the source point ID and uses it to retrieve current attributes values, including tag name, from the source point.



Caution: If the UserInt1 attribute is used for other purposes, only use negative values to prevent misinterpretation as a source point ID. Consequently, the PItoPI_APS Connector is not able to synchronize an interface point if the source point is renamed.

Limited Support of Custom Point Classes

The architecture of APS requires an APS Connector to provide a list of supported attributes *without* connecting to a data source or obtaining information from it. Therefore, an APS Connector cannot dynamically adapt to what it finds in a data source. In the case of the PItoPI_APS Connector, it is designed to support the attributes in the table in the [Supported PI Point Attributes](#) section, which includes most of the attributes from the Classic point class. When the PItoPI_APS Connector needs to retrieve attributes for a source PI Data Archive point, it examines the point class to determine what attributes to request from the source point. If the point class includes the Classic attribute set, the PItoPI_APS Connector requests all supported attributes. If the point class does not include the Classic attribute set, the PItoPI_APS Connector only requests the supported attributes from the Base point class. Therefore, only the supported Base class attributes are used from a source point in a point class that does not include the Classic attribute set.

When the PItoPI_APS Connector retrieves attributes for a source point in a point class that does *not* include the Classic attribute set, the handling of the missing Classic attributes depends on whether the PItoPI_APS Connector was called to find available points for creation or to get current source attributes for synchronizing existing interface points.

- If the PItoPI_APS Connector was called for available points, the missing Classic attributes are set to built-in values shown in the “Default Value” column in the table in the [Supported PI Point Attributes](#) section.
- If the PItoPI_APS Connector was called to get current source point attributes for existing target points, no value is returned for the missing Classic attributes, which the PI APS Synchronization Engine treats as unchanged attributes. That is, the missing Classic attributes are not synchronized.

Source PI Data Archive-level Failover Restrictions

The PItoPI Interface supports a *Source PI Data Archive-level Failover* feature. This feature designates a secondary source PI Data Archive that the interface can use when the primary source PI Data Archive is unavailable. As a historical note, the source PI Data Archive-failover feature preceded the development of PI Data Archive collectives. Prior to PI Data Archive collectives, the primary and secondary source PI Data Archives were independent PI Data Archives. Therefore, corresponding points with the same name in the two source PI Data Archives are not guaranteed to have the same point ID.

The PItoPI_APS Connector has no corresponding capability. The PItoPI_APS Connector always synchronizes with the primary source PI Data Archive. If the primary source PI Data Archive is unavailable at the time a synchronization scan occurs, the synchronization scan fails and the APS Synchronization Engine tries again at the next scheduled synchronization interval.

The [How PItoPI and PItoPI_APS Find Source Points](#) section explains how the interface point key attributes are used to find the source point for an interface point.

In the most general and most common case, four key attributes are examined as possible links to the source point: InstrumentTag, ExDesc, UserInt1, and Tag, in that order. The UserInt1 attribute is the only key attribute that is interpreted as a source point ID, which is significant when source PI Data Archive-level failover is configured. The other three key attributes can contain only the source point tag name.

When the interface is configured with a secondary source PI Data Archive, the PItoPI Interface rejects an interface point that is mapped to a source point by source point ID in the UserInt1 attribute. The reason for this restriction is that corresponding points with the same name in the two source PI Data Archives are not guaranteed to have the same point ID. For source PI Data Archive-level failover to work correctly, the interface needs the point IDs for both source points in the interface point configuration but the UserInt1 attribute can hold only one. Therefore, when the interface is configured for source PI Data Archive-level failover, interface points are prohibited from mapping to source points by using the UserInt1 attribute as source point ID.

The PItoPI_APS Connector supports synchronizing the interface point attributes that contain source point tag name when the source point is renamed. To make this possible, the PItoPI_APS Connector provides an option to store the source point ID in the UserInt1 attribute value of interface points. The option to store the source point ID in the UserInt1 attribute value of interface points can be enabled even if the interface is configured to ignore UserInt1 as a key attribute. Since the PItoPI_APS Connector only synchronizes with the primary source PI Data Archive, the point IDs that the PItoPI_APS Connector optionally returns for the UserInt1 attribute value are from the primary source PI Data Archive.

Unless the interface is configured to ignore UserInt1 as a key attribute or the interface finds a source point tag name in the InstrumentTag or ExDesc attributes, the interface considers the UserInt1 attribute as the point ID for mapping the source point. When source PI Data

Archive-level failover is enabled, the interface rejects an interface point with source point ID in the UserInt1 attribute. Therefore, the PItoPI_APS Connector cannot provide the source point ID for the UserInt1 attribute value of an interface point unless the interface is configured to ignore UserInt1 as a key attribute or the interface point is mapped to the source point by a tag name in the InstrumentTag or ExDesc attributes. Without the source point ID in the UserInt1 attribute, the PItoPI_APS Connector cannot locate a renamed source point.

Note: When source PI Data Archive-level failover is enabled, the PItoPI_APS Connector can synchronize an interface point with a renamed source point only if the interface is configured to ignore UserInt1 as a key attribute or the original source point tag name is in the InstrumentTag or ExDesc attributes, not the Tag attribute.

PI2 Data Archives Not Supported

The PI SDK does not support PI2 Data Archives. Therefore, the PItoPI_APS Connector cannot access a source PI2 Data Archive.

Source Point Tag Name Length Limit in PItoPI

The PItoPI Interface uses PI API to communicate with the target PI Data Archive. PI API imposes an 80-character limit on the length of a tag name. Therefore, the interface cannot map by tag name to a source point that has a tag name longer than 80 characters. If a source point has a tag name longer than 80 characters, the source point ID must be stored in the UserInt1 attribute of the interface point.

If an interface point is mapped by source point ID to a source point with a tag name longer than 80 characters, the PItoPI_APS Connector can synchronize the interface point. If the PItoPI_APS Connector is configured to store source point tag name in InstrumentTag or ExDesc, and a source point tag name is longer than 80 characters, the tag name is not stored in attributes that the interface actively uses as key attributes.

If the PItoPI_APS Connector finds a source point that is available for creation, the source point tag name is longer than 80 characters, and the following conditions are satisfied, PItoPI_APS can construct an available point that (if created) is mapped to the source point by ID and, therefore, loadable by the interface:

- The interface actively uses UserInt1 as a key attribute.
- The PItoPI_APS Connector option to store source point ID in UserInt1 is selected.

If the PItoPI_APS Connector is configured to store source point tag name in InstrumentTag or ExDesc, and a source point tag name is longer than 80 characters, the tag name is not stored in attributes that the interface actively uses as key attributes.

Chapter 2. Principles of Operation

This section explains how the PItoPI_APS Connector and APS Synchronization Engine work together to synchronize PI points for an instance of the PItoPI Interface.

The APS Synchronization Engine runs as a Windows service. It appears in the Windows Task Manager on the **Processes** tab as `PIAPSEngine.exe`. The APS Synchronization Engine schedules synchronization scans and implements the synchronization tasks common to all APS Connectors. The APS Synchronization Engine makes all PI SDK calls to obtain or change attributes for existing points or to create new interface points. The APS Synchronization Engine does not access the source PI Data Archive directly.

The PItoPI_APS Connector implements a specific set of functions required by the APS Synchronization Engine. It is an in-process COM object that is implemented in a dynamic-link library (DLL) named `PItoPI_APS.dll`. This object is registered during installation of the PItoPI_APS Connector. The APS Synchronization Engine dynamically loads the PItoPI_APS Connector. That is, the PItoPI_APS Connector “plugs into” the APS Synchronization Engine.

During a synchronization scan for the PItoPI Interface, the APS Synchronization Engine calls functions in the PItoPI_APS Connector that actually obtain source point definitions by calling the PI SDK to access the source PI Data Archive. The PItoPI_APS Connector returns source point information from the source PI Data Archive to the APS Synchronization Engine. The PItoPI_APS Connector does not access or change the *target* PI point database.

The PItoPI_APS Connector has two primary functions that the APS Synchronization Engine calls:

- Acquire a list of *available points*, which are points in the source PI Data Archive without a corresponding interface point, and
- Obtain current source PI Data Archive attributes for *existing points* for the interface.

If APS is configured for automatic operation, the APS Synchronization Engine uses information returned by the PItoPI_APS Connector to synchronize existing PItoPI Interface instance points in the target PI Data Archive, create new interface points, and remove existing points when corresponding source points are deleted. Alternatively, the APS Synchronization Engine stores the results of the last synchronization scan in a database file. The APS Configuration Utility can read the results of the last synchronization scan from the database file and display the pending edits for existing points, existing points that can be deleted, and new points that can be created. Also, the APS Configuration Utility allows you to selectively apply pending edits, create new points, or delete points.

Point Class

This APS Connector returns *Classic* when APS asks for the point class to use when creating new interface points. APS creates only Classic points for an interface instance registered with the PItoPI_APS Connector. Existing Classic points can be edited or deleted.

If the interface instance has existing points in a point class other than Classic and the point class contains all of the attributes listed in the table in the [Supported PI Point Attributes](#) section, the APS Synchronization Engine can obtain updated attributes and detect deleted points for the non-Classic points. However, if an existing point belongs to a class that does *not* contain all attributes in the table, the point cannot be synchronized because errors result when the APS Synchronization Engine requests the current attribute values for attributes that are not defined in the class. Therefore, use the APS Configuration Utility to disable synchronization for any point in a class that does not contain all of the attributes supported by the PItoPI_APS Connector.

Point Source and Instance ID

PI points are associated with a PItoPI Interface instance by the PointSource and Location1 attributes. APS obtains the PointSource and instance ID values for an interface instance from its ICU configuration. APS uses these attributes and the values from ICU to identify the existing points for an interface instance. If APS creates a point for the interface instance, the values from ICU are assigned to the PointSource and Location1 attributes of the new points.

The PItoPI Interface supports a **/c1** parameter that causes the interface to ignore the Location1 attribute while loading existing points. APS implements a corresponding special case while identifying the existing points for a PItoPI Interface instance with a **/c1** parameter. Also, when the **/c1** parameter is present, the Location1 attribute from source points provides the Location1 attribute value for creating or synchronizing interface points.

Key Attributes

The PItoPI_APS Connector uses the following PI point attributes to identify and access the source point for an interface point:

- InstrumentTag
- ExDesc
- UserInt1
- Tag

The PItoPI Interface uses the same attributes to identify and access the source point for an interface point.

Depending on PItoPI Interface parameters, some of the key attributes may be ignored by both the interface and the PItoPI_APS Connector.

How PItOPI and PItOPI_APS Find Source Points

When the PItOPI Interface loads a point, the interface must identify the source point from which data will be collected. In the *PI to PI Interface User Guide* and other parts of this user guide, this process is referred to as *mapping* the interface point to a source point.

The PItOPI Interface has three parameters that affect the procedure for finding the source point: **/tn**, **/tnex**, and **/ptid**. For most interface instances, none of these parameters are used and any of the key attributes can contain the mapping to the source point. The interface parameters cause the PItOPI Interface and the PItOPI_APS Connector to ignore one or more of the key attributes.

The following table summarizes the effect of these parameters on the search for either source point tag name or ID in the key attributes. The actual implementation of the search is described below the table.

Interface parameters			Search order for attribute containing source point tag name or ID
/tn	/tnex	/ptid	
No	No	No	1. Non-empty InstrumentTag attribute is source point tag name. 2. ExDesc attribute that begins with STAG= specifies source point tag name. 3. UserInt1 attribute greater than 0 contains source point ID. (Point rejected if source PI Data Archive-level failover configured.) 4. Tag of the interface point is also source point tag name.
No	No	Yes	UserInt1 attribute contains source point ID. (Illegal configuration if source PI Data Archive-level failover is configured.)
No	Yes	No	1. ExDesc attribute that begins with STAG= specifies source point tag name. 2. Tag of the interface point is also source point tag name.
No	Yes	Yes	Tag of the interface point is also source point tag name.
Yes	N/A	N/A	

PItoPI Interface Procedure to Identify Source Point

The interface performs the following steps to find the source point:

1. If no **/tn** parameter and no **/tnex** parameter and no **/ptid** parameter and the InstrumentTag attribute is not a zero-length string, the InstrumentTag attribute contains the source point tag name and the search ends. Otherwise, proceed to step 2.
2. If no **/tn** parameter and no **/ptid** parameter and the ExDesc attribute begins with case-insensitive STAG followed by zero or more spaces followed by =, the source point tag name is extracted from the remainder of the ExDesc attribute (as described in the following paragraph) and the search ends. Otherwise, proceed to step 3.

To extract the source point tag name, the interface ignores any spaces following the =. If the next character is not a double quotation mark ("), it is the first character of the source point tag name which extends to the first comma or to the end of the ExDesc attribute. If the first non-space following the = is a double quotation mark, the source point tag name begins with the following character and extends to the first double quotation mark or to the end of the ExDesc attribute. The interface extracts the source point tag name from ExDesc and the search ends.

3. If no **/tn** parameter and no **/tnex** parameter and either the UserInt1 attribute is greater than zero or the **/ptid** parameter is present, the UserInt1 attribute is the source point ID and the search ends. Otherwise, proceed to step 4. If the search ends in this step and either the UserInt1 attribute is not greater than zero or the interface is configured for source PI Data Archive-level failover, the interface rejects the point and the PItoPI_APS Connector returns *no updates available* status for the existing point.
4. The source point tag name is the same as the interface tag name.

The search for the source point ends with either a source point tag name or ID. If the source point tag name or ID does not exist, the interface puts the point into an error state and logs a message.

PItoPI_APS Connector Extensions

The PItoPI_APS Connector follows the same procedure as the interface to identify either a source point tag name or ID. If the search ends with a source point tag name and the source point tag name does not exist, then the source point is either renamed or deleted. To distinguish between these two possibilities, the PItoPI_APS Connector needs the source point ID. The PItoPI_APS Connector provides a configuration option to store source point ID in the UserInt1 attribute for this purpose.

If the option to store source point ID in UserInt1 is not selected, the PItoPI_APS Connector does not assume that UserInt1 contains the source point ID. Without the source point ID, the PItoPI_APS Connector cannot determine if the source point is renamed and, therefore, must indicate to the APS Synchronization Engine that the corresponding source point is deleted. Consequently, the APS Synchronization Engine applies the delete rule to the interface point. If the source point is actually renamed, the PItoPI_APS Connector finds and returns the renamed source point when searching for available points. The APS Synchronization Engine applies the create rule to the renamed source point.

Recommendation: Select the option to store source point ID in UserInt1 and ensure that synchronization of UserInt1 is enabled for all interface points.

If the option to store source point ID in UserInt1 is selected, the PItoPI_APS Connector tests whether the UserInt1 attribute value is a valid point ID on the source PI Data Archive.

If a source point with the point ID does not exist, the PItoPI_APS Connector indicates to the APS Synchronization Engine that the corresponding source point is deleted. Consequently, the APS Synchronization Engine applies the delete rule to the interface point.

If a source point exists with the point ID, the source point is renamed. The PItoPI_APS Connector provides current attribute values from the source point to the APS Synchronization Engine for synchronizing the interface point. Specifically, the PItoPI_APS Connector replaces the source point tag name in the interface point attributes where the old source point tag name was stored. Since the interface point Tag attribute is either the source point tag name or a name constructed from the source point tag name, the interface point is renamed. Also, the new source point tag name is stored in the InstrumentTag or ExDesc attributes that are selected for storing the source point tag name.

Creatable and Synchronizable Attributes

The PItoPI_APS Connector supports the PI point attributes listed in the table in the [Supported PI Point Attributes](#) section. The *Sync.* column indicates whether an attribute is used for point creation only or for both point creation and synchronization of existing points.

When the APS Synchronization Engine calls the APS Connector for available points, the APS Connector attempts to provide a value for every attribute in the table.

Note: Per-point synchronization settings do not apply when APS creates a PI point.

When the APS Synchronization Engine calls the APS Connector for current source PI Data Archive attribute values for existing points, the APS Connector attempts to provide a value only for the attributes with *Yes* in the *Sync.* column. The APS Connector does not provide values for the attributes with *No* in the *Sync.* column.

Available and Hidden Points

The APS Synchronization Engine initially categorizes source points that the APS Connector discovers without a corresponding point in the target PI Data Archive as *available points*. Using the APS Configuration Utility, available points can be selectively changed into *hidden points* and hidden points can be changed back into available points. Both available and hidden points correspond to source PI Data Archive points with no associated interface points. The APS Synchronization Engine handles the two categories differently: available points are available for creation in the target PI Data Archive, and hidden points are treated as if they do not exist.

During each synchronization scan, the APS Synchronization Engine constructs an array containing the key attributes of existing interface points and hidden points for the interface instance.

The APS Synchronization Engine passes the array of key attributes for existing and hidden points to the PItoPI_APS Connector in a call for available points.

The PItoPI_APS Connector queries the source PI Data Archive for a list of its point definitions by calling the PI SDK GetPoints or GetPointsSQL method. Both methods have a

WHERE clause parameter and the methods return a list of source PI Data Archive points that satisfy the conditions specified in the WHERE clause. The WHERE clause is configurable and the default WHERE clause is `Tag = '*'` which returns all points on the source PI Data Archive. The choice of query method is configurable and the default method is `GetPoints`. The [Source Point Filter](#) section discusses configuration of the query method and WHERE clause.

Using the key attributes, the PItoPI_APS Connector matches the points in the array of existing and hidden points from the APS Synchronization Engine with the points in the list from the source PI Data Archive. Points in the source PI Data Archive that *match* a point in the array of existing and hidden points are *not* available points.

Unmatched points in the source PI Data Archive are returned to the APS Synchronization Engine as available points for point creation.

Tag Naming Conventions

By default, available points are assigned the same tag name as the corresponding point on the source PI Data Archive. However, the PItoPI_APS Connector provides a configurable pattern that constructs tag names from a combination of the source point tag name, literal text, and optionally the interface instance name. The [Tag Naming Convention](#) section discusses configuration of the pattern for tag name construction.

If the option to store source point ID in UserInt1 is selected, the PItoPI_APS Connector is able to rename an interface point when the source point is renamed.

Point Type

An available point is assigned the same point type as the source PI Data Archive point.

Pause Between Groups

To reduce resource usage while matching the points in the array of existing and hidden points from the APS Synchronization Engine with the points in the list from the source PI Data Archive, the PItoPI_APS Connector periodically pauses. The main effect of this feature is on CPU usage by APS. Since the source points that satisfy the WHERE clause are retrieved by a single call to the source PI Data Archive, this feature generally does not reduce the number of calls to the source PI Data Archive. However, if the array of existing and hidden points contains a significant number of points that are not in the list from the source PI Data Archive, this feature also may reduce the rate to calls to the source PI Data Archive.

This feature is controlled by two general APS parameters: the group size for *available* points and the amount of time to pause between groups (see *Interface Point Group Sizes* and *Milliseconds to Pause between Interface Point Group Sizes* in the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide). After the PItoPI_APS Connector matches *n* existing and hidden points, where *n* is the available point group size, the PItoPI_APS Connector suspends itself for the configured time to pause between groups.

Updated Attributes

During each synchronization scan, the APS Synchronization Engine constructs an array containing the key attributes of existing PI points for the interface instance that are enabled for synchronization.

The APS Synchronization Engine passes the array of synchronizable existing points to the PItoPI_APS Connector in a call for updated attributes.

For each synchronizable existing point, the PItoPI_APS Connector uses the key attributes to map to a source PI Data Archive point and read the current attribute values. The APS Connector assembles the results for each existing point into an array to return to the APS Synchronization Engine.

If the key attributes identify a valid source PI Data Archive point, the APS Connector adds the current attribute values from the source point to the array of information for return to the APS Synchronization Engine.

If the key attributes do not identify a valid source PI Data Archive point, the APS Connector adds an indication that the existing point has no point in the source PI Data Archive to the array for return to the APS Synchronization Engine.

Note: The PItoPI_APS Connector indicates no source point if and only if the existing interface point does not map to a valid source PI Data Archive point. As long as the source PI Data Archive point exists, even if the source PI Data Archive point does not satisfy the WHERE clause for selecting available points, the APS Synchronization Engine will *not* apply the rule for deleted points to an existing point. See the [Decommissioned Source PI Data Archive Points](#) section.

After the APS Connector processes all of the synchronizable existing points, the array of results is returned to the APS Synchronization Engine. The APS Synchronization Engine uses the information in the array to generate a list of interface points to update and a list of interface points to delete.

Chapter 3. Installation, Upgrading, and Uninstallation Instructions

Installation Instructions

APS must be installed on the same computer as the PItoPI_APS Connector.

PI AutoPointSync version 1.2.5.0 or later is required.

The PItoPI_APS Connector setup program uses the services of the Microsoft Windows Installer. Windows Installer is a standard part of Windows 2000 and later.

To install, run the `PItoPI_APS_#. #. #. #. #. exe` installation kit.

Note: This installation kit installs only the PItoPI_APS Connector, not the PItoPI Interface or APS.

Installation Checklist

1. Log on as a user with Administrator privileges.
2. PI AutoPointSync version 1.2.5.0 or later is required. If version 1.2.5.0 or later is not already installed, install or upgrade PI AutoPointSync. (The APS installation kit also contains the PI SDK, PI API, and PI ICU. If necessary, the APS installation kit will install or *upgrade* these components.)



Caution: Installing APS can *upgrade* PI API and PI SDK, which may stop operational interfaces on the computer. If installation stops any interfaces, the interfaces must be *manually* restarted.

3. Run `PItoPI_APS_#. #. #. #. #. exe` to install the PItoPI_APS Connector.

Upgrading Instructions

For information on the current version of the PItoPI_APS Connector, see the [OSIsoft Customer Portal](#).

The same `PItoPI_APS_#. #. #. #. #. exe` installation kit is used for either an initial installation or upgrading an existing installation of the PItoPI_APS Connector.

Upgrading Checklist

1. PI AutoPointSync version 1.2.5.0 or later is required. If version 1.2.5.0 or later is not already installed, observe the caution notice in the [Installation Checklist](#) section before upgrading APS.
2. Log on as a user with Administrator privileges, preferably the same user who originally installed the PItoPI_APS Connector.
3. If the APS Configuration Utility is running, exit from the program.
4. Stop the APS Synchronization Engine. The most common ways of stopping the APS Synchronization Engine are from the Services utility or from a command window. For convenience, the Services utility or command window can be left open and used later to restart the APS Synchronization Engine.

If using the Services utility, scroll to **PI APS Synchronization Engine** in the right pane and click on that item to select it. Then, click **Stop** on the **Actions** menu or toolbar.

If using a command window, enter:

```
net stop piapsengine
```

5. Run the installation kit.
6. Restart the APS Synchronization Engine.

If using the Services utility, click **Start** on the **Actions** menu or toolbar.

If using a command window, enter:

```
net start piapsengine
```


Uninstallation Instructions

To uninstall the PItoPI_APS Connector:

1. Review the interface instances that are registered with APS. Unregister all interface instances that are associated with the PItoPI_APS Connector.
2. If the APS Configuration Utility is running, exit from the program.
3. Log on as a user with Administrator privileges, preferably the same user who originally installed the PItoPI_APS Connector.
4. Stop the APS Synchronization Engine. The most common ways of stopping the APS Synchronization Engine are from the Services utility or from a command window. For convenience, the Services utility or command window can be left open and used later to restart the APS Synchronization Engine.

If using the Services utility, scroll to **PI APS Synchronization Engine** in the right pane and click on that item to select it. Then, click **Stop** on the **Actions** menu or toolbar.

If using a command window, enter:

```
net stop piapseengine
```

5. Depending on the version of Windows:
Click **Start > Settings > Control Panel > Add or Remove Programs** to open the **Add or Remove Programs** utility.
—or—
Click **Start > Control Panel > Uninstall a program** to open the **Uninstall or change a program** utility.
6. Remove the following program:
 - o PI to PI Interface - APS Connector
7. Restart the APS Synchronization Engine.

If using the Services utility, click **Start** on the **Actions** menu or toolbar.

If using a command window, enter:

```
net start piapseengine
```


Chapter 4. Registering an Interface Instance with APS

PI AutoPointSync synchronizes the points for interface *instances* that are registered with APS. When an interface has multiple instances, registering one instance with APS synchronizes only the points for that specific instance. Each interface instance must be registered with APS for its points to be synchronized.

If you are not familiar with registering interface instances with APS, this section does *not* provide the in-depth discussions of the configurable options that are essential for successful configuration. Important concepts and detailed explanations of the configurable options are presented in the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide, which you must read before proceeding with the steps in this chapter. The [Connector-specific Configuration Control](#) chapter in this user guide is also prerequisite reading. The [Principles of Operation](#) chapter is strongly recommended reading.

Note: OSIsoft emphasizes the need to read both user guides before beginning to register the first interface with APS.

If you are familiar with APS, the procedures in this section give the main steps for registering an interface instance. These procedures are presented in the order that you must perform them.

1. [Register the Interface](#)
2. [Configure the Settings](#)
3. [Enable Synchronization](#)

Register the Interface

Only interface instances managed by the Interface Configuration Utility can be registered with APS. If a new interface instance is being created or an existing interface instance is not managed by ICU, run ICU and add the interface instance. Refer to the *PI Interface Configuration Utility* and the *PI to PI Interface User Guide* user guides for instructions.

Recommendation: Assign a meaningful name to the interface instance instead of accepting the default generic name.

Most interface instances are configured with a single PointSource string. If an interface instance is configured with multiple PointSource strings, however, the first string in the ICU **Point Source** list is the default PointSource attribute value for points created by APS. Confirm that the first PointSource string in the ICU **Point Source** list is the appropriate default for points created by APS.

Configure PI SDK Connection to Target PI Data Archive

In order for APS to synchronize points on a PI Data Archive, you must configure PI SDK to connect to the PI Data Archive and configure the PI Data Archive to allow the connections from APS programs to access securable objects.

1. Run the APS Configuration Utility.
2. Click **SDK Connections** on the **Interface** menu. Confirm that the server list on the **PI Connections Manager** dialog box contains the target PI Data Archive with the interface points that you want to synchronize. If not, use the **PI Connections Manager** dialog box to add the target PI Data Archive.
3. Configure PI Trusts or PI Mappings on the target PI Data Archive that allow the APS Synchronization Engine and APS Synchronization Trigger service to automatically connect. Optionally, configure a PI Trust or PI Mapping for the APS Configuration Utility. See the section *Required PI Server Configuration* in the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide and the PI Data Archive documentation for more information.
4. Optionally, configure APS to use only specific PI Data Archives in the PI SDK Known Servers Table. The Known Servers Table is a system-wide resource that is shared by all PI SDK applications on the computer where APS runs. The default APS configuration causes APS to poll all servers in the Known Servers Table for interface instances registered with APS. However, PI Data Archives in the Known Servers Table for other applications might not contain interface instances registered with APS on this computer. Polling these PI Data Archives is unnecessary overhead for both the PI Data Archives and APS. The *Loading Tab* section in the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide explains the options for limiting the PI Data Archives that APS polls for interface instances to synchronize.

Configure PI SDK Connection to Source PI Data Archive

In order for the PItoPI_APS Connector to obtain point information from the source PI Data Archive, you must configure PI SDK to connect to the source PI Data Archive and configure the source PI Data Archive to allow the connections from APS programs to access securable objects.

Because the PItoPI_APS Connector runs in the APS Synchronization Engine program, the source PI Data Archive must grant the access permissions in the following table to the APS Synchronization Engine.

PI Securable Object	Access Permission
PIPOINT table	R
Individual PI Points (PtAccess or PtSecurity attribute)	R
PIDS table	R

The connector-specific configuration control has a **Test Expression** button that tests the WHERE clause for retrieving points from the source PI Data Archive. To use the **Test Expression** button, the source PI Data Archive must grant the access permissions in the preceding table to the APS Configuration Utility (because the connector-specific configuration control runs in the APS Configuration Utility). Otherwise, the APS Configuration Utility does not access the source PI Data Archive.

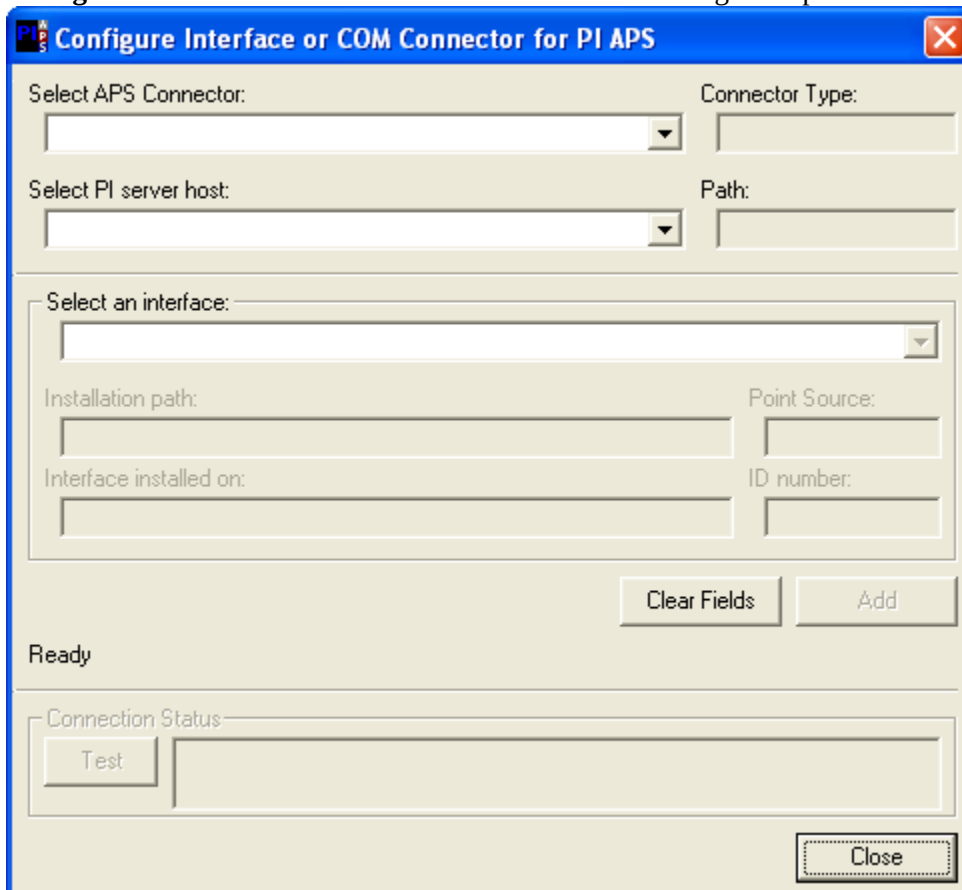
The APS Synchronization Trigger service does not access the source PI Data Archive.

Generally, an APS node that synchronizes interface instances for the target PI Data Archive will not have registered interface instances for the source PI Data Archive. So, polling the source PI Data Archive is unnecessary overhead. Also, if the network connection to the source PI Data Archive crosses a slow communication link (like a wide-area network), APS Synchronization Engine performance can be adversely affected.

1. Configure APS to use only specific PI Data Archives in the PI SDK Known Servers Table and exclude the source PI Data Archive. The *Loading Tab* section in the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide explains the options for limiting the PI Data Archives that APS polls for interface instances to synchronize.
2. In the APS Configuration Utility, click **SDK Connections** on the **Interface** menu. Confirm that the server list on the **PI Connections Manager** dialog box contains the *source* PI Data Archive for the interface points that you want to synchronize. If not, use the **PI Connections Manager** dialog box to add the source PI Data Archive.
3. Configure a PI Trust or PI Mapping on the source PI Data Archive that allows the APS Synchronization Engine to automatically connect. Optionally, configure a PI Trust or PI Mapping for the APS Configuration Utility. Use the access permissions in the preceding table to configure the PI Trusts or PI Mappings. See the PI Data Archive documentation for more information on configuring security.

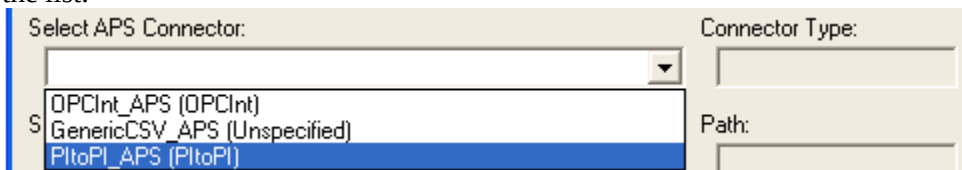
Register the PItoPI Interface

1. In the APS Configuration Utility, click **Register New...** on the **Interface** menu. The **Configure Interface or COM Connector for PI APS** dialog box opens:



For details on using the **Configure Interface or COM Connector for PI APS** dialog box, see the *Configure Interface or PI COM Connector for PI APS Dialog Box* section in the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide.

2. Click the **Select APS Connector** arrow and select the **PItoPI_APS** Connector from the list:



3. Click the **Select PI server host** arrow and select the target PI Data Archive where points for the interface instance either exist or will be created.

After making a selection in both boxes, the APS Configuration Utility loads the **Select an interface** list with the interface instances that are not yet registered with APS.

4. Click the **Select an interface** arrow and select a specific interface instance for registration.
5. Click **Add**.

Configure the Settings

Click the arrow in the **Interface** box and select the interface instance to configure.

Configure the APS options for the new interface instance on the dialog boxes that are opened by the following commands on the **Settings** menu:

- **Rules...**
- **Sync Schedule...**
- **User-set Defaults...**
- **Connector-specific...**

Review the settings on these dialog boxes because registration presets synchronization rules to non-automatic selections and the default per-point synchronization settings to disable synchronization of each point. The preset options are designed to prevent APS from changing the PI point database until explicitly enabled. Therefore, it is likely that the default settings need to be changed for individual interface instances.

Note: OSIsoft emphasizes the necessity to configure these options *before* enabling a new interface instance. In particular, options on the **User-set Defaults** dialog box establish the default per-point synchronization settings that are assigned to existing points only once on the *first* synchronization scan. Therefore, these settings must be configured for your specific needs *prior to the first synchronization scan*.

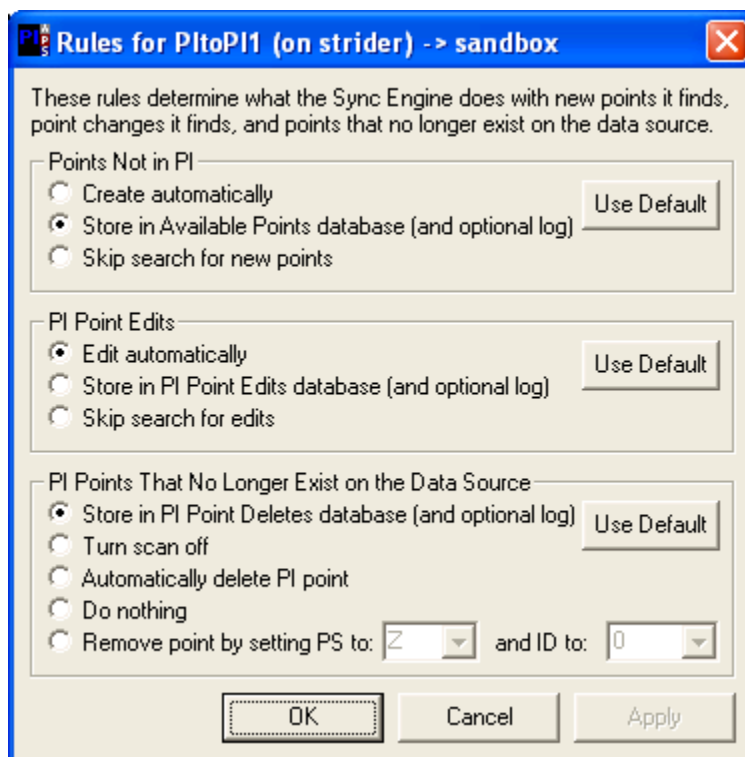
The settings dialog boxes are briefly described in these sections:

- [Rules](#)
- [Synchronization Schedule](#)
- [User-set Defaults](#)
- [Connector-specific Configuration Control](#)

During initial registration, you do not need to review or configure the options on the dialog boxes opened by the **Sync Settings...**, **Debug...**, **Columns...**, and **Column Header Names...** commands on the **Settings** menu.

Rules

On the **Settings** menu, click **Rules**. The **Rules for interface** dialog box opens:



The options on this dialog box determine how the APS Synchronization Engine handles available points, attributes of existing points that disagree with current values in the source PI Data Archive, and existing points that are not configured for valid source points. For a full explanation of this dialog box, see section *Rules Dialog Box* in the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide.

Note: OSIsoft emphasizes that you should initially choose the options to store the synchronization results. Carefully review the results of several synchronization scans for expected behavior before selecting any options to automatically update the PI point database.

The APS Synchronization Engine detects interface points that are not configured for valid source points while scanning for attributes that need to be edited. Therefore, if the **Skip search for edits** option is selected, interface points that are candidates for deletion are not detected. When the **Skip search for edits** option is selected, the options in the **PI Points That No Longer Exist on the Data Source** area have no effect and the APS Synchronization Engine behaves like the **Do nothing** option is selected.

Note: An existing interface point is *not* a candidate for deletion unless the interface point does not map to a valid source PI Data Archive point. That is, decommissioning a source PI Data Archive point by any means other than actually deleting the source point does not make the existing interface point a candidate for deletion. See the [Decommissioned Source PI Data Archive Points](#) section.

Synchronization Schedule

On the **Settings** menu, click **Sync Schedule**. The **Synchronization Schedule for interface** dialog box opens:

Synchronization Schedule for PItoPI1 (on strider) -> sandbox

Synchronization Schedule

The Synchronization Schedule determines when the PI AutoPointSync Sync Engine synchronizes this interface's points.

☒ Synchronize automatically using the following schedule:

Days: Hours: Minutes:

☐ Do not synchronize automatically. Synchronization is started by clicking 'Sync Now' on the toolbar or by defining a Synchronization Trigger.

Synchronization Trigger

Sync Trigger Configuration

Name:

Type:

Trigger On:

Attribute:

Sync once on startup: ☐

Configured Sync Triggers:

Name	Type	TriggerOn	Attribute

This dialog box configures:

- Automatic scheduling of synchronization scans by the APS Synchronization Engine.
- Events that automatically initiate a synchronization scan.

For a full explanation of this dialog box, see section *Synchronization Schedule Dialog Box* in the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide.

Note: Even though this dialog box configures automatic synchronization scans, the handling of differences between interface points and source points is still subject to the settings on the **Rules** dialog box. The rules might not be automatic.

The **Synchronization Schedule** area configures whether the APS Synchronization Engine initiates synchronization scans for the interface instance on a periodic schedule. If **Synchronize automatically using the following schedule** is selected, the **Period** boxes specify how often the APS Synchronization Engine runs a synchronization scan for the interface instance.

Note: If you select **Synchronize automatically**, the first synchronization starts immediately when the interface instance is enabled.

The **Do not synchronize automatically** button prevents periodic scheduling of synchronization scans for the interface instance. When this button is selected, synchronization scans occur only when initiated by clicking **Sync Now** on the toolbar or by the APS Synchronization Trigger service.

The options in the **Synchronization Trigger** area configure and display the conditions that the APS Synchronization Trigger service monitors. When the condition for a trigger occurs, the APS Synchronization Trigger service notifies the APS Synchronization Engine that a synchronization scan is needed.

Note: Unless periodic scheduling or synchronization triggers are configured, you must click **Sync Now** on the APS Configuration Utility toolbar to start a synchronization scan.

User-set Defaults

On the **Settings** menu, click **User-set Defaults**. The **User-set Defaults for interface** dialog box opens. The **User-set Defaults** dialog box contains tabs that switch between two sets of options. The **Tag Naming** and **Tag Selection** tabs are unavailable because the PItoPI_APS Connector does not support these features.

For a full explanation of the **User-set Defaults for interface** dialog box, see section *User-set Defaults Dialog Box* in the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide.

Security & Archive Settings

On the **User-set Defaults for *interface*** dialog box, click the **Security & Archive Settings** tab:

The screenshot shows the 'User-set Defaults for PItoPI1 (on strider) -> sandbox' dialog box with the 'Security & Archive Settings' tab selected. The dialog has four tabs: 'Security & Archive Settings', 'Initial Sync Masks', 'Tag Naming', and 'Tag Selection'. The 'Security & Archive Settings' tab is active and contains three main sections: 'Point & Data Default Security Settings', 'Compression & Exception Default Settings', and 'Default Settings'.

Point & Data Default Security Settings

- ☒ Use old security model attributes.
 - Data Access: o:rw | g:r | w:r (Buttons: Undo, PI Default)
 - Data Group: piadmins (Dropdown)
 - Data Owner: piadmin (Dropdown)
 - Point Access: o:rw | g:r | w:r (Buttons: Undo, PI Default)
 - Point Group: piadmins (Dropdown)
 - Point Owner: piadmin (Dropdown)
- ☐ Use new security model attributes.
 - Data Security: piadmin: A(r,w) | piadmins: A(r,w) | PIWorld: A(r) (Button: Build)
 - Point Security: piadmin: A(r,w) | piadmins: A(r,w) | PIWorld: A(r) (Button: Build)

If the APS Connector does not support these attributes, PI APS uses the values above when it creates new PI points.

Compression & Exception Default Settings

- Compressing: On (Buttons: Undo, PI Default)
- Compdev: 2 % of Span (Buttons: Undo, PI Default)
- Excdev: 1 % of Span (Buttons: Undo, PI Default)
- Compmax: 28800 (Buttons: Undo, PI Default)
- Excmax: 600 (Buttons: Undo, PI Default)
- Compmin: 0 (Buttons: Undo, PI Default)
- Excmin: 0 (Buttons: Undo, PI Default)

If the APS Connector does not support these attributes, PI APS uses the values above when it creates new PI points.

Default Settings

- Shutdown: Off (Buttons: Undo, PI Default)

If the APS Connector does not support these attributes, PI APS uses the values above when it creates new PI points.

Buttons at the bottom: OK, Cancel, Apply.

When APS creates a new interface point, the attribute default values from this tab are assigned to any attributes for which the PItoPI_APS Connector did not provide a value. The **Shutdown** default value is never used because the PItoPI_APS Connector always provides the source point Shutdown attribute value. For the other attributes on this tab, the PItoPI_APS Connector can be configured to either provide source point attribute values or no values. In the latter configuration, APS assigns the default attribute values from this tab. See [Point Attributes](#) for more information.

The attribute default values from this tab are *not* used to synchronize existing points.

Initial Sync Masks

On the **User-set Defaults for interface** dialog box, click the **Initial Sync Masks** tab:

The screenshot shows a dialog box titled "User-set Defaults for PItoPI1 (on strider) -> sandbox". It has four tabs: "Security & Archive Settings", "Initial Sync Masks" (selected), "Tag Naming", and "Tag Selection".

The "Initial Sync Masks" tab is divided into two sections: "Existing PI Points" and "Available Points".

Existing PI Points:

Sync	PI Attribute
☐	archiving
☒	compdev
☒	compdevpercent
☒	compmax
☒	compmin
☒	compressing
☐	convers

Point is sync'd by APS:
☐ Yes
☒ No

Buttons: Use Default, Check All, Uncheck All

When PI APS discovers an existing PI point without per-point sync settings, PI APS assigns the settings in this area to the PI point. Changing these settings does not affect PI points that already have per-point sync settings.

Available Points:

Sync	PI Attribute
☐	archiving
☒	compdev
☒	compdevpercent
☒	compmax
☒	compmin
☒	compressing
☐	convers

Point is sync'd by APS:
☐ Yes
☒ No

Buttons: Use Default, Check All, Uncheck All

When PI APS creates a new PI point, PI APS assigns the per-point sync settings in this area to the PI point.

At the bottom are buttons: OK, Cancel, Apply.

When APS encounters an existing point for the first time and the existing point does not have per-point synchronization settings, APS assigns the default settings in the **Existing PI Points** area to the point. The settings in this area are only used in the following situations:

- On the first synchronization scan for an interface instance that has existing points when the interface instance is registered with APS.
- If the APS Configuration Utility loads an existing point into the grid area of the main window and the existing point does not have per-point synchronization settings.
- After points are added to an interface instance by tools other than APS.

After synchronization settings are assigned to an existing point, changes made to the settings in the **Existing PI Points** area do not affect the per-point synchronization settings that were originally assigned.

The APS Synchronization Engine assigns the per-point synchronization settings in the **Available Points** area to interface points that it creates automatically or the APS Configuration Utility assigns these settings to interface points that it creates under user control.

Note: The per-point synchronization settings assigned to created points control the attributes that can be edited on subsequent synchronization scans. *All* attributes from the APS Connector are used for point creation. That is, the per-point synchronization settings do *not* apply to point creation.

Both areas contain a list of the synchronizable attributes supported by the APS Connector and two options under **Point is sync'd by APS**. The **Point is sync'd by APS:** options select the master synchronization setting assigned to a point. When an interface is registered, the initial **Point is sync'd by APS:** selections default to **No**. These defaults were chosen to ensure that the APS Synchronization Engine does not synchronize any points until explicitly enabled. The master synchronization setting for a point must be **Yes** for the APS Synchronization Engine to synchronize any attributes.

Note: The **Point is sync'd by APS** options determine the master synchronization setting for a point. The default setting is **No**, which prevents the APS Synchronization Engine from synchronizing the point.

The check boxes in the lists select the individual synchronization setting for each attribute. The initial settings for the individual attribute synchronization options are obtained from the APS Connector. The settings for individual attributes can be toggled by clicking the check boxes in the list.

Considerations for Renamed Source Points

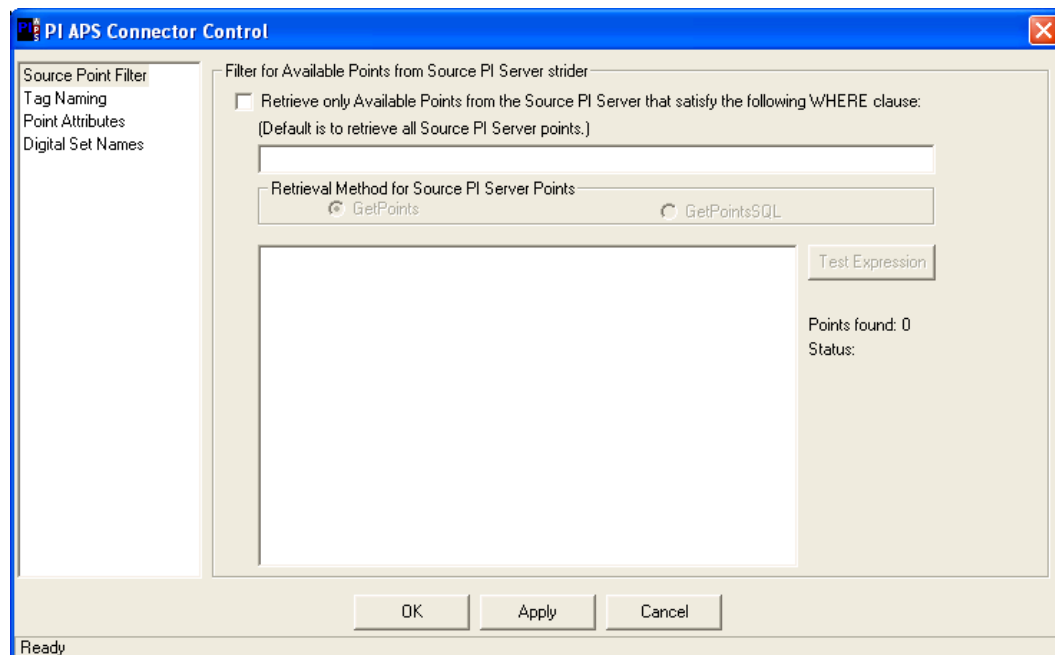
For the PItOPI_APS Connector to synchronize an interface point whose source point has been renamed, the UserInt1 attribute must be set to the point ID of the source point before renaming occurs. To prepare for possible renaming of source points, the PItOPI_APS Connector provides an option to store the source point ID as the value for the UserInt1 attribute. If synchronization is enabled for the UserInt1 attribute, APS stores the source point ID in the UserInt1 attribute. OSIsoft recommends selecting the option to store source point ID in UserInt1 and enabling synchronization of the UserInt1 attribute.

Depending on interface configuration (see the [How PItOPI and PItOPI_APS Find Source Points](#) section), the source point tag name can be configured in the InstrumentTag, ExDesc, or Tag attribute of the interface point. For the PItOPI_APS Connector to propagate the new name of a renamed source point into the interface point, synchronization must be enabled on the attributes that contain source point tag name. Observe that the Tag attribute is not in the lists. The NEWTag attribute actually controls synchronization of the interface point tag name.

Recommendation: Enable synchronization for the key attributes that the PItOPI_APS Connector is configured to set to source point tag name or ID.

Connector-specific Options

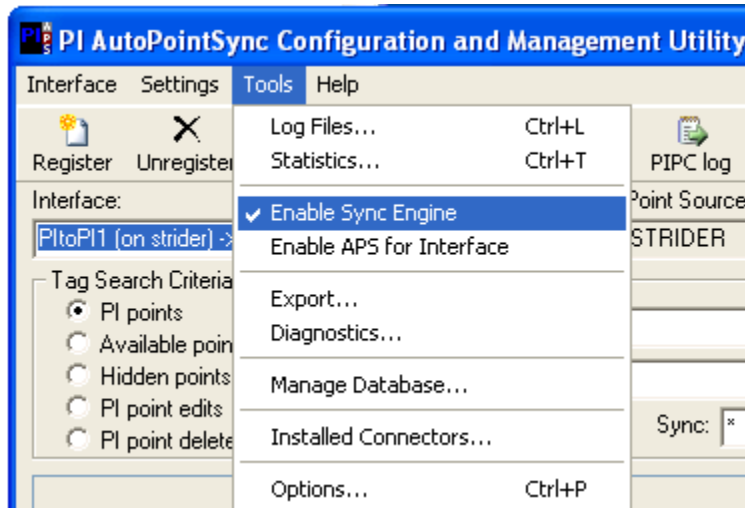
The PItoPI_APS Connector has additional configuration options. On the **Settings** menu, click **Connector-specific** to open the **PI APS Connector Control** dialog box:



The [Connector-specific Configuration Control](#) chapter in this user guide provides a detailed discussion of the options on this dialog box.

Enable Synchronization

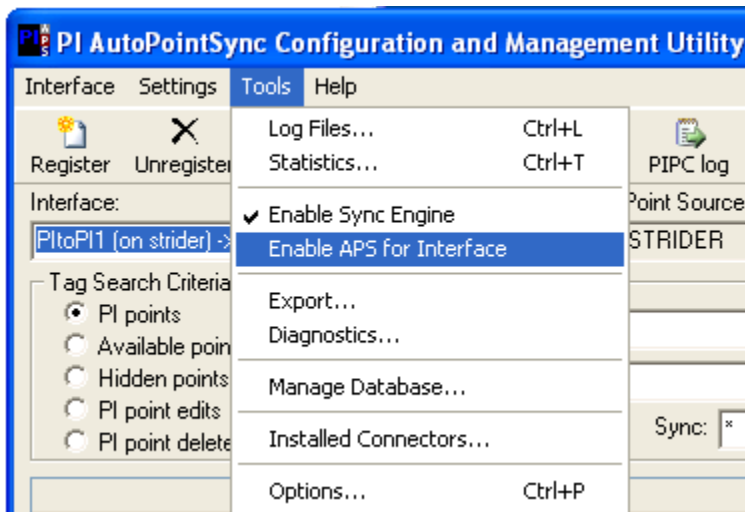
1. The APS Synchronization Engine is installed in disabled mode and must be enabled before any interface instances can be synchronized. Click the **Tools** menu. If the **Enable Sync Engine** command has a check mark next to it as shown in the following figure, then the APS Synchronization Engine is enabled and no action is required. Otherwise, click **Enable Sync Engine** to enable the APS Synchronization Engine.



2. When an interface instance is registered with APS, the interface instance is in disabled mode and must be enabled before it can be synchronized.

Note: If an interface instance is configured for automatic scheduled synchronization, the first synchronization scan starts immediately when the APS Connector is enabled for the interface instance. Confirm that configurations of Rules, Sync Schedule, and User-set Defaults are appropriate for your needs *before* enabling the APS Connector.

Click the **Tools** menu. If no check mark appears by the **Enable APS for Interface** command as shown in the following figure, click **Enable APS for Interface** to enable synchronization for the interface instance.

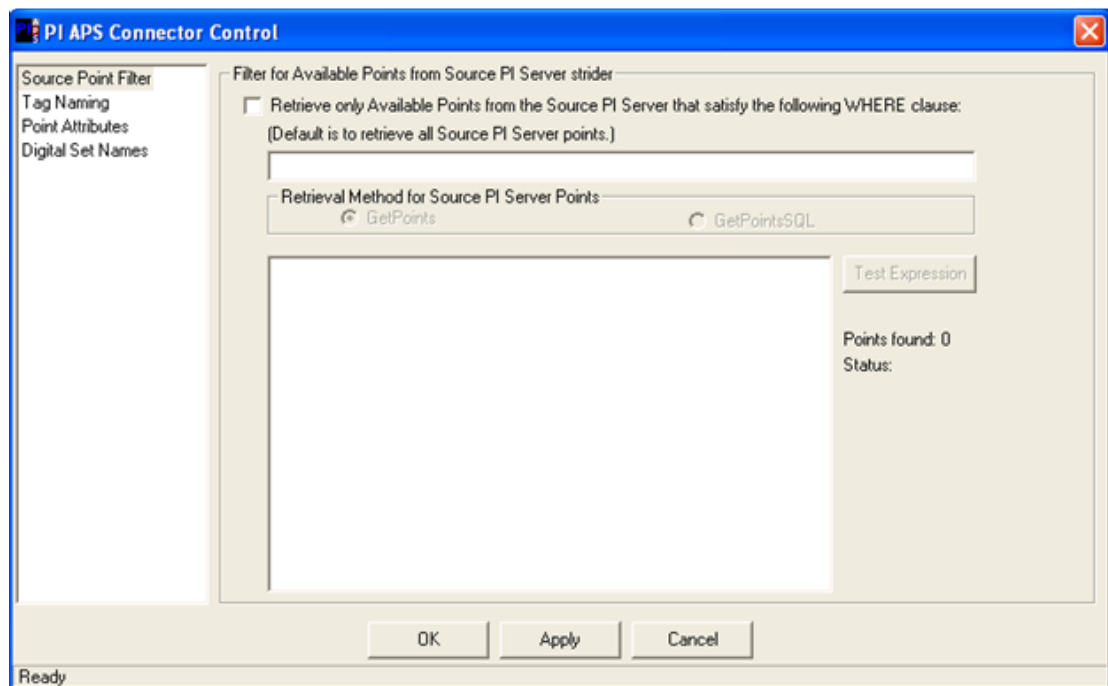


Chapter 5. Connector-specific Configuration Control

A connector-specific configuration control is installed with the PItoPI_APS Connector. This control configures options that affect operation of the PItoPI_APS Connector.

- Filter for available source PI Data Archive points.
- Attributes where source point tag name or ID is stored and Tag naming pattern.
- Security settings for available points when they are created in PI Data Archive.
- Archive settings for available points when they are created in PI Data Archive.
- Whether to keep the SourceTag attribute from the source point.
- Scan class for available points when they are created in PI Data Archive.
- Digital Set Names.

Access the configuration control from the APS Configuration Utility. In the APS Configuration Utility, select the interface instance to configure. On the **Settings** menu, click **Connector-specific...** to open a dialog box that contains the connector-specific control.



The list on the left selects groups of configurable options. Click an item in the list to show a specific group of options on the right side of the dialog box.

Source Point Filter

The connector-specific configuration control opens with **Source Point Filter** selected in the list on the left as shown in the preceding figure. Or, click **Source Point Filter** in the list on the left to show the options related to filtering the source PI Data Archive points considered for *available* points.

The source PI Data Archive that this interface instance uses is shown in the label at the top of the right pane.

Note: The **Source Point Filter** does *not* apply to *existing* interface points. If an existing interface point maps to a valid source PI Data Archive point, even if the source PI Data Archive point does not satisfy the **Source Point Filter**, the rule for deleted points is *not* applied to the existing interface point. See the [Decommissioned Source PI Data Archive Points](#) section.

If the **Retrieve only Available Points from the Source PI Server that satisfy the following WHERE clause** check box is cleared, all points on the source PI Data Archive are retrieved as candidate available points.

When the **Retrieve only Available Points from the Source PI Server that satisfy the following WHERE clause** check box is selected, enter a WHERE clause in the box. The PtoPI_APS Connector calls the PI SDK GetPoints or GetPointsSQL method with the WHERE clause as the query string parameter. The source PI Data Archive returns a list of points that satisfy the query. The PtoPI_APS Connector then considers this list of points to belong to this instance of the PtoPI Interface.

Note: When the WHERE clause is changed, the available points from the last synchronization scan using the *old* WHERE clause remain in the APSPoints database for the interface instance. Therefore, after changing the WHERE clause, OSIsoft recommends that you purge available points from the APSPoints database (**Manage Databases...** on the **Tools** menu) and review any hidden points.

The **GetPoints** and **GetPointsSQL** buttons select which query method is used. The GetPoints and GetPointsSQL methods have different capabilities and performance characteristics that influence the choice of method. GetPoints is generally faster than GetPointsSQL but GetPoints cannot handle complex queries.

Note: Some WHERE clause attribute names are different for GetPoints and GetPointsSQL. GetPoints uses the native attribute names of the PI3 Point Database and GetPointsSQL uses the attribute names defined by PI ODBC.

The PI SDK on-line help provides information about the GetPoints and GetPointsSQL methods. In the PI SDK on-line help, these methods are in the *PI SDK Programming Reference* under the Server object. The GetPoints2 method under the IGetPoints2 interface contains a *GetPoints vs. GetPointsSQL* section that summarizes the differences between GetPoints and GetPointsSQL.

The configuration control has a **Test Expression** button that allows the WHERE clause to be tested before it is saved. Click **Test Expression** to call the selected query method with the WHERE clause. The first 200 points returned by the query are displayed in the box on the left of the **Test Expression** button. The number of points returned by the query and the return status of the query are displayed under the **Test Expression** button.

GetPoints WHERE clause syntax summary

The following description of WHERE clause syntax for the GetPoints method is summarized from the on-line help for PI SDK version 1.3.8. Since the PI SDK can be updated independently of APS, the WHERE clause summary in this document is not a substitute for PI SDK on-line help, which is the authoritative reference.

- A WHERE clause (query expression) is composed of one or more *phrases* separated by **or**.
- A *phrase* is composed of one or more *basic tests* separated by **and**.
- A *basic test* is:
attribute-name relational-operator value
 The relational operators are: = (equal), <> (not equal), <, <=, >, and >=.

When relational operator equal (=) or not equal (<>) is used with a string-valued attribute, *value* can contain “*” or “?” wildcard characters. When *value* contains wildcard characters, *value* is a pattern that the attribute value must match for the equal operator to be true (or not match for the not equal operator to be true).

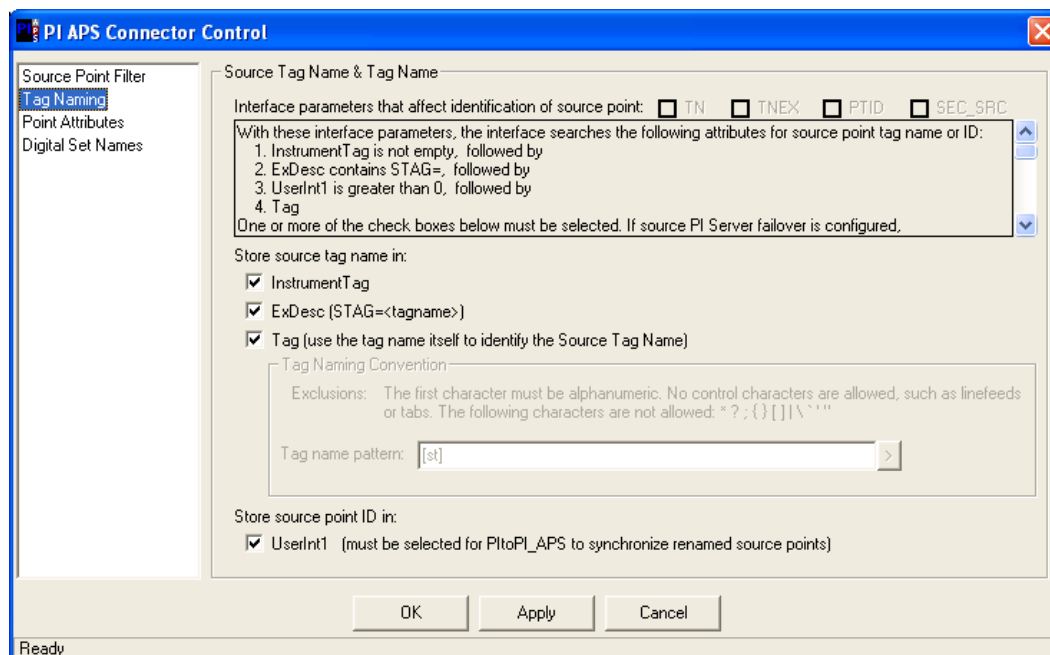
If a phrase is composed of more than one basic test, an attribute name cannot be used more than once in the phrase. For example,
span > 10 and span < 50
 is an illegal phrase because *span* is used more than once.

Support for parentheses is limited to enclosing entire phrases. That is, parentheses cannot contain an **or** operator. For example,
tag = 'sin*' and (pointsource = 'L' or pointsource = 'R')
 is an illegal WHERE clause. However, this expression can be rewritten as
(tag = 'sin*' and pointsource = 'L') or (tag = 'sin*' and pointsource = 'R')
 which is a legal WHERE clause composed of two phrases.

When a WHERE clause is composed of more than one phrase, the PI SDK sends each phrase as a separate query to the source PI Data Archive. If a point satisfies more than one phrase, the source PI Data Archive returns the point multiple times. If a large number of points satisfy more than one phrase, this can increase the load on the source PI Data Archive to process the WHERE clause and on the network to transfer the same point multiple times. Other than increased load, a multi-phrase WHERE clause has no other negative consequences. The PI SDK merges the individual lists from the multiple phrases into a single, final point list that is returned to the client application. Before adding a point to the final list for the client application, the PI SDK checks that the point is not already in the final list. Thus, the final list does not contain duplicate points.

Tag Naming

Click **Tag Naming** in the list on the left to show the options to select the interface point attributes where the source point tag name or ID is configured and to specify how interface point tag names are constructed.



The PItoPI Interface has three parameters that affect the process of identifying the source point: `/tn`, `/tnex`, and `/ptid`. For most interface instances, none of these parameters are used. Depending on the specific interface parameters, the interface searches for source point tag name in zero or more of the InstrumentTag, ExDesc, or Tag attributes of an interface point. Unless the `/tn` or `/tnex` parameter is present or the interface is configured for source PI Data Archive-level failover, the PItoPI Interface considers the UserInt1 attribute as a possible source point ID. If the interface is configured for source PI Data Archive-level failover (`/sec_src` parameter), see the [Source PI Data Archive-level Failover Restrictions](#) section for additional information and restrictions.

The upper portion of the **Source Tag Name & Tag Name** area contains indicators for the presence or absence of the `/tn`, `/tnex`, `/ptid`, or `/sec_src` parameters in the interface configuration. For the specific interface parameters in effect, the quick reference box under the parameter indicators shows the sequence of key attributes that the interface searches to identify the source point.

Note: The PItoPI Interface and the PItoPI_APS Connector expect that all existing points are properly configured by the active key attributes which are determined by the interface parameters.

The following table summarizes the effect of the interface parameters on the key attributes that are searched for either source point tag name or ID. The actual implementation of the procedure for finding the source point is described in the [How PItoPI and PItoPI_APS Find Source Points](#) section.

Interface parameters			Search order for attribute containing source point tag name or ID
/tn	/tnex	/ptid	
No	No	No	1. Non-empty InstrumentTag attribute is source point tag name. 2. ExDesc attribute that begins with STAG= specifies source point tag name. 3. UserInt1 attribute greater than 0 contains source point ID. (Point rejected if source PI Data Archive-level failover configured.) 4. Tag of the interface point is also source point tag name.
No	No	Yes	UserInt1 attribute contains source point ID. (Illegal parameter if source PI Data Archive-level failover is configured.)
No	Yes	No	1. ExDesc attribute that begins with STAG= specifies source point tag name. 2. Tag of the interface point is also source point tag name.
No	Yes	Yes	Tag of the interface point is also source point tag name.
Yes	N/A	N/A	

The **Source Tag Name & Tag Name** area contains a check box for each of the key attributes that the interface can search for source point tag name or ID. In conjunction with the interface parameters, these check boxes configure the value that the PItoPI_APS Connector provides for the corresponding interface point attributes. In this user guide, the discussion of each check box lists the values that the PItoPI_APS Connector returns as a function of the check box state and interface parameters. For the specific interface parameters in effect, the quick reference box on the control summarizes the values that the PItoPI_APS Connector returns for each state of the check boxes for the key attributes that can contain the source point tag name or ID.

Note: The quick reference box does not include all of the details and special cases that are discussed in this section.

Generally, the PItoPI_APS Connector returns either

- the source point tag name or ID (check box selected), or
- the corresponding source point attribute (check box cleared).

However, if the check box for a key attribute is cleared and the interface is configured to search the attribute for the source point tag name or ID, the PItoPI_APS Connector cannot return the corresponding source point attribute value in the following cases:

- If the value can be mistaken for a source point tag name or ID, the PItoPI_APS Connector either modifies the value in a way that prevents misinterpretation or provides a value that cannot be mistaken for a source point tag name or ID.
- If changing the key attribute value for an existing point removes the only link to the source point, the PItoPI_APS Connector either does not update the key attribute, or, if the attribute contains source point tag name and the source point is renamed, replaces the source point tag name in the attribute.

The second case may seem overly restrictive but is necessary because the PItoPI_APS Connector does not have access to per-point synchronization settings for existing points.

Consequently, the PItoPI_APS Connector does not know which, if any, attribute edits will actually occur. Therefore, when the PItoPI_APS Connector chooses values to return for key attributes, the PItoPI_APS Connector must assume the worst case: changing a key attribute to map to a source point does not occur and changing a key attribute to remove a map to a source point does occur. With these assumptions, the PItoPI_APS Connector cannot change a key attribute to remove a map if the key attribute is the only key attribute that maps to the source point.

A check box must be selected for at least one of the key attributes that the interface searches for source point tag name or ID. Selecting more than one is optional.

For key attributes that the interface does not search because of interface parameters, selecting the corresponding check box is optional. The source point tag name or ID can be stored in the ignored key attributes for documentation purposes or in preparation for changing the interface configuration to use an attribute in the search sequence. If the check box is not selected for a key attribute that the interface ignores for identifying source point, the PItoPI_APS Connector provides:

- the corresponding source point attribute value for interface point InstrumentTag, ExDesc, or UserInt1 attributes.
- a name constructed from the **Tag name pattern** for the interface point Tag attribute.

As explained in the [PItoPI_APS Connector Extensions](#) section, the PItoPI_APS Connector requires source point ID to detect a renamed source point. Therefore, even if the interface does not search the UserInt1 attribute for source point ID, the option to store source point ID in UserInt1 is recommended.

InstrumentTag

The PItoPI_APS Connector uses the state of this check box and the interface parameters to select the value to provide for the InstrumentTag attribute of an interface point according to the following table.

Check box	No /tn, /tnex, or /ptid parameter in the PItoPI Interface configuration (normal configuration)	/tn, /tnex, or /ptid parameter in the PItoPI Interface configuration
	In this configuration, the PItoPI Interface checks the InstrumentTag attribute for the source point tag name. Therefore, the InstrumentTag attribute must either contain the source point tag name or be empty.	In this configuration, the PItoPI Interface ignores the InstrumentTag attribute, which allows it to be synchronized with the InstrumentTag attribute of the source PI Data Archive point.
Cleared 	Available point: The PItoPI_APS Connector provides an empty string for the InstrumentTag attribute value. Existing point: Unless the existing InstrumentTag value is the only link to the source point, the PItoPI_APS Connector provides an empty string for the new InstrumentTag attribute value. If the existing InstrumentTag value is the only link to the source point, the source point tag name is the new InstrumentTag value.	The PItoPI_APS Connector provides the source point InstrumentTag attribute for the InstrumentTag attribute value.
Selected 	The PItoPI_APS Connector provides the source point tag name for the InstrumentTag attribute value.	The PItoPI_APS Connector provides the source point tag name for the InstrumentTag attribute value.

If the **InstrumentTag** check box is selected, the per-point synchronization settings also must enable InstrumentTag synchronization for APS to store source point tag name in the InstrumentTag attribute of an existing interface point.

ExDesc

The PtoPI_APS Connector uses the state of this check box and the interface parameters to select the value to provide for the ExDesc attribute of an interface point according to the following table.

Check box	No /tn or /ptid parameter in the PtoPI Interface configuration	/tn or /ptid parameter in the PtoPI Interface configuration
	In this configuration, the PtoPI Interface checks the ExDesc attribute for a source point tag name specification. If the ExDesc attribute begins with STAG=tag , tag is the source point tag name. Otherwise, the ExDesc attribute is ignored.	In this configuration, the PtoPI Interface ignores the ExDesc attribute, which allows it to be synchronized with the ExDesc attribute of the source PI Data Archive point.
Cleared 	Available point: The PtoPI_APS Connector provides the source point ExDesc attribute for the ExDesc attribute value, subject to Note 1. Existing point: Unless the existing ExDesc value contains a source point tag name specification that is the only link to the source point, the PtoPI_APS Connector provides the source point ExDesc attribute for the ExDesc attribute value, subject to Note 1. If the existing ExDesc value contains a source point tag name specification that is the only link to the source point, the PtoPI_APS Connector provides STAG=source_tag_name for the new ExDesc attribute value.	The PtoPI_APS Connector provides the source point ExDesc attribute for the ExDesc attribute value.
Selected 	The PtoPI_APS Connector provides STAG=source_tag_name for the ExDesc attribute value.	The PtoPI_APS Connector provides STAG=source_tag_name for the ExDesc attribute value.

Note 1: If the source point ExDesc attribute value begins with a source point tag name specification **STAG=tag**, tag is almost certain to reference some point other than the source point. Therefore, the ExDesc attribute value must be altered to prevent the interface from interpreting it as a source point tag name specification. The PtoPI_APS Connector prefixes an underscore character to the ExDesc attribute value in this case.

If the **ExDesc** check box is selected, the per-point synchronization settings also must enable ExDesc synchronization for APS to store source point tag name in the ExDesc attribute of an existing interface point.

Tag

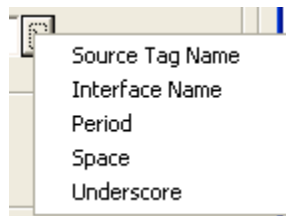
The PItoPI_APS Connector uses the state of this check box and the interface parameters to select the value to provide for the Tag attribute of an interface point according to the following table. If the interface parameters contain **/tn** or both **/tnex** and **/ptid**, the interface point tag name is the only attribute used for source point tag name; therefore, the configuration control automatically selects this check box and prevents it from being cleared.

Check box	/tn parameter, or /tnex parameter, or no /ptid parameter in the PItoPI Interface configuration	/ptid parameter and neither of the /tn or /tnex parameters in the PItoPI Interface configuration
	In this configuration, the PItoPI Interface may use the Tag attribute for the source point tag name.	In this configuration, the PItoPI Interface ignores the Tag attribute, which allows it to be synchronized with the tag name constructed from the Tag name pattern .
Cleared 	Available point: The PItoPI_APS Connector provides a tag name constructed from the Tag name pattern . Existing point: Unless Tag is the only key attribute that maps to the source point, the PItoPI_APS Connector provides a tag name constructed from the Tag name pattern . If Tag is the only key attribute that maps to the source point, the PItoPI_APS Connector provides the source point tag name.	The PItoPI_APS Connector provides a tag name constructed from the Tag name pattern .
Selected 	The PItoPI_APS Connector provides the source point tag name.	The PItoPI_APS Connector provides the source point tag name.

Note: For APS to rename an existing point, the NEWTag attribute must be enabled for synchronization.

Tag Naming Convention

If the **Tag** check box is cleared, then the **Tag name pattern** box specifies a pattern for constructing interface point tag names. The default tag name pattern is the source point tag name. The  button opens the following menu of options that may be used as part of the tag name pattern:



- **Source Tag Name** (name of the source point)
- **Interface Name** (name of the interface these points belong to)
- **Period** (“.”)
- **Space** (“ ”)
- **Underscore** (“_”)

In addition, the tag name pattern may include any other character that the PI Data Archive accepts as a valid tag name character. The illegal characters for PI tag names are listed on the control.

The source point tag name must be included somewhere in the **Tag name pattern** box.

If an illegal character is entered in the box or the source point tag name is missing, the box color changes to yellow, and the illegal pattern will not be saved.

UserInt1

The PItoPI_APS Connector uses the state of this check box and the interface parameters to select the value to provide for the UserInt1 attribute of an interface point according to the following table

Check box	No /tn or /tnex parameter in the PItoPI Interface configuration	/tn or /tnex parameter in the PItoPI Interface configuration
	In this configuration, the /ptid parameter determines the PItoPI Interface search sequence. Without /ptid , the interface only checks the UserInt1 attribute if none of InstrumentTag and ExDesc contain the source point tag name. With /ptid , the interface requires source point ID in UserInt1. The check box must be selected.	In this configuration, the PItoPI Interface ignores the UserInt1 attribute.
Cleared 	Available point: The PItoPI_APS Connector provides the source point UserInt1 attribute, subject to Note 1. Existing point: Unless the existing UserInt1 value contains a source point ID that is the only link to the source point, the PItoPI_APS Connector provides the source point UserInt1 attribute, subject to Note 1. If the existing UserInt1 value contains a source point ID that is the only link to the source point, the PItoPI_APS Connector does not update UserInt1.	The PItoPI_APS Connector provides the source point UserInt1 attribute.
Selected 	The PItoPI_APS Connector provides the source point ID, subject to Note 2.	The PItoPI_APS Connector provides the source point ID.

Note 1: If a source point UserInt1 attribute value greater than zero is stored in UserInt1 of an interface point and the search sequence reaches the UserInt1 attribute, the interface interprets the value as a point ID. To prevent the interface from interpreting UserInt1 as a source point ID, the PItoPI_APS Connector replaces a source point UserInt1 attribute value greater than zero with 0.

Note 2: If the interface is configured for source PI Data Archive-level failover, the source point tag name must be stored in the InstrumentTag or ExDesc attribute to prevent the search sequence from considering UserInt1. If not, the interface refuses to load an existing point with UserInt1 greater than zero. If storing source point ID in UserInt1 results in a point that the interface would refuse:

- For an available point, the PItoPI_APS Connector writes a message to the synchronization log file and does not return the available point to the APS Synchronization Engine.
- For an existing point, the PItoPI_APS Connector does not provide a value for UserInt1, which prevents the APS Synchronization Engine from changing UserInt1.

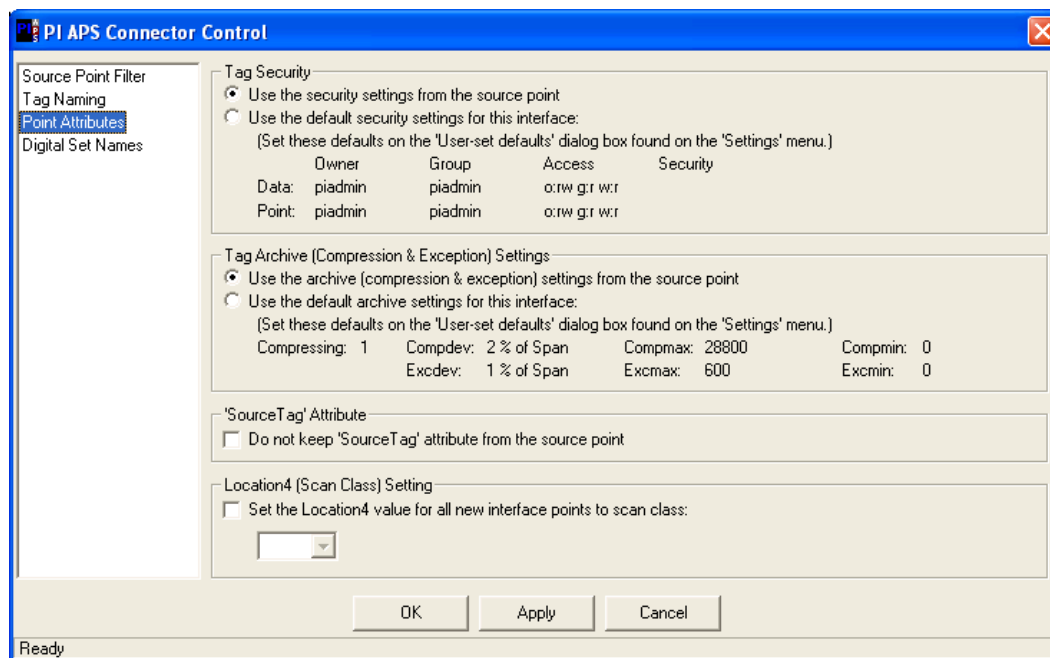
If the **UserInt1** check box is selected, the per-point synchronization settings also must enable UserInt1 synchronization for APS to store source point tag name in the UserInt1 attribute of an existing interface point.

As explained in the [PItoPI_APS Connector Extensions](#) section, the PItoPI_APS Connector requires source point ID to detect a renamed source point. Therefore, even if the interface does not search the UserInt1 attribute for source point ID, the option to store source point ID in UserInt1 is recommended.

Note: The option to store source point ID in UserInt1 must be selected for the PItoPI_APS Connector to detect a renamed source point.

Point Attributes

Click **Point Attributes** in the list on the left to show the configuration options for point attributes.



Tag Security

If the **Use the security settings from the source point** option is selected, then the security attributes from the source PI Data Archive point are used by APS when creating the point on the target PI Data Archive. If the source point DataGroup, DataOwner, DataSecurity, PtGroup, PtOwner, or PtSecurity attributes contain a user, group, or identity that does not exist on the target PI Data Archive, point creation fails.

Note: APS does not add users, groups, or identities to the target PI Data Archive.

If the **Use the default security settings for this interface** option is selected, the security attributes from the source point are not used when creating the interface point. Instead, the default values specified on the **User-set defaults** dialog box are used.

The **Tag Security** options do not affect synchronization of existing points. The PItoPI_APS Connector always returns the current source point security attribute values to the APS Synchronization Engine. Use the per-point synchronization settings to inhibit APS from updating interface point attributes from the source point.

Tag Archive (Compression & Exception) Settings

If the **Use the archive (compression & exception) settings from the source point** option is selected, then the exception and compression attributes from the source PI Data Archive point are used by APS when creating the point on the target PI Data Archive.

If the **Use the default archive settings for this interface** option is selected, the exception and compression attributes from the source point are not used when creating the interface point. Instead, the default values specified on the **User-set defaults** dialog box are used.

The **Tag Archive (Compression & Exception) Settings** do not affect synchronization of existing points. The PItoPI_APS Connector always returns the current source point archive attribute values to the APS Synchronization Engine. Use the per-point synchronization settings to inhibit APS from updating interface point attributes from the source point.

'SourceTag' Attribute

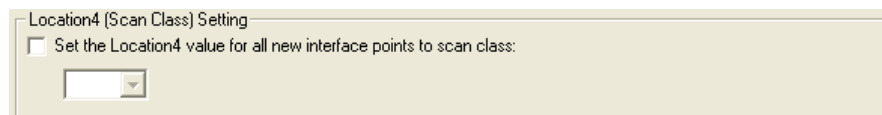
If the **Do not keep 'SourceTag' attribute from the source point** check box is selected, the SourceTag attribute from the source point is not used when creating an available point. If this check box is not selected and the tag name specified in the SourceTag attribute does not exist on the target PI Data Archive, point creation fails.

The **Do not keep 'SourceTag' attribute from the source point** check box does not affect synchronization of existing points. The PItoPI_APS Connector always returns the current source point SourceTag attribute value to the APS Synchronization Engine. Use the per-point synchronization settings to inhibit APS from updating interface point attributes from the source point.

Location4 (Scan Class) Setting

If the **Set the Location4 value for all new interface points to scan class** check box is not selected, the default scan class for new (available) points is 1.

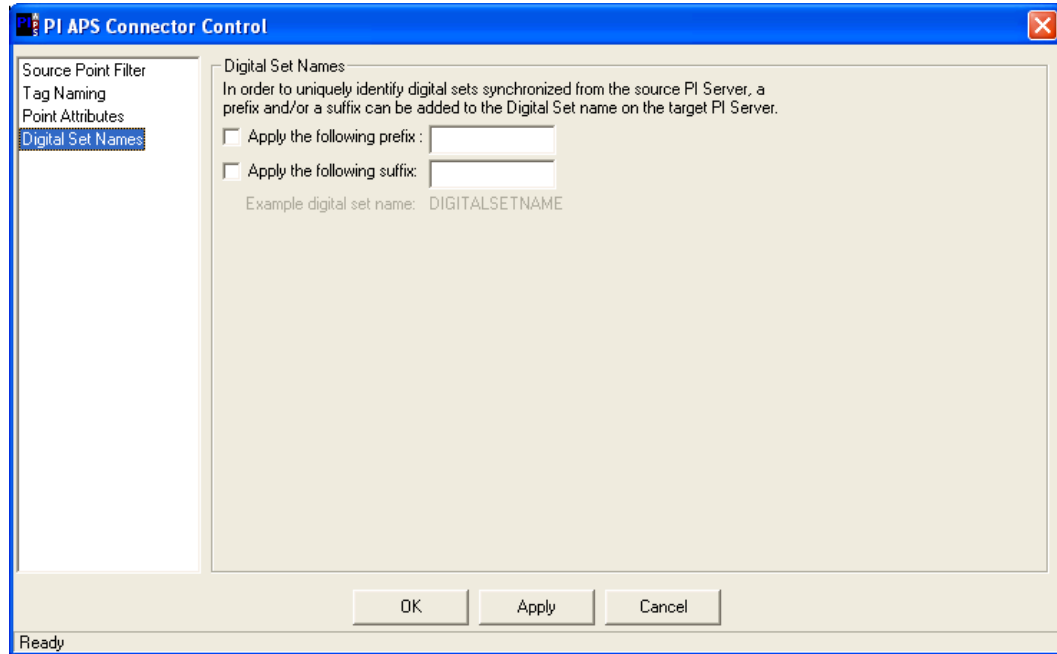
If you select the **Set the Location4 value for all new interface points to scan class** check box, the PItoPI_APS Connector configuration control allows you to specify a scan class (Location4 value) for all new interface points. The configuration control loads all scan classes defined for the PItoPI Interface instance being configured into the list where you can select a scan class number:



If the interface instance has a **/c4=x** command line parameter, this option is unavailable. The PItoPI_APS Connector provides the source point Location4 attribute value as the value for the target point Location4 attribute.

Digital Set Names

Click **Digital Set Names** in the list on the left to show the configuration options for digital set names.



The configuration options in the **Digital Set Names** area add a prefix, suffix, or both to digital set names from the source PI Data Archive before using them on the target PI Data Archive. By using these options, the digital sets for the points of this interface instance can be separated from the digital sets for other interfaces to the target PI Data Archive. For example, if source PI Data Archive and target PI Data Archive have digital sets with the same name, the two digital sets with the same name may not contain the same list of states. This may be a temporary condition because one PI Data Archive is changed before the other or a permanent condition. In either case, the interface points for this interface instance need the state list from the source PI Data Archive and all other points on the target PI Data Archive need the list of states on the target server, which is impossible with a common digital set name. The only solution is to maintain separate digital sets for the PItOPI Interface instance by using these options.

A common application for the PItOPI Interface is to consolidate points from PI Data Archives at several remote sites onto a central PI Data Archive. Each PI Data Archive may have its own administrator. The administrator of the central PI Data Archive and the PItOPI Interfaces may have little or no control over the remote sites. The remote sites may create new digital sets or change the states in existing digital sets without coordinating with either the central site or other remote sites. In this situation, maintaining separate digital sets for the PItOPI Interface instances is the only practical alternative for managing digital sets on the central PI Data Archive.

Chapter 6. Best Practices, Hints, and Techniques

This chapter addresses special cases and describes how to accomplish some of the less common or more specialized tasks with APS for the PtoPI Interface.

PtoPI Interface with Multiple Instances

APS assumes that it manages *all* points in the source PI Data Archive on behalf of a PtoPI Interface registered with APS. When multiple instances of the PtoPI Interface are collecting data from one source PI Data Archive, special care must be taken when registering these interfaces with APS to avoid the problems discussed in the *Interfaces with Multiple Instances and PI APS* section of the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide. Refer to the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide for a complete explanation of the problems and the two main configuration methods for registering multiple interfaces from one source PI Data Archive with APS. The discussion below is a summary of the two configuration methods.

On a target PI Data Archive where multiple instances of the PtoPI Interface are collecting data from one source PI Data Archive, APS assigns new interface points on the source PI Data Archive to each instance of the interface that it synchronizes. If APS is configured to create the points automatically, the create operation succeeds for the first PtoPI Interface instance that is synchronized after a new source point is created. When APS synchronizes the other PtoPI Interface instances, APS also detects the new source point. If the APS instances are configured with the same tag naming pattern, the create operation fails because the interface point created for the first PtoPI instance has already taken the tag name. If the APS instances are configured with different tag naming patterns, the create operation succeeds but the target PI Data Archive will have multiple (redundant) points for the new source point.

Another issue with registering multiple interfaces for one source PI Data Archive with APS is that the existing interface points for one PtoPI Interface instance appear to be new points to the other PtoPI Interface instances. As with actual new points on the source PI Data Archive, either point creation fails (resulting in error messages in the log files) or redundant points are created.

The following two methods can be used to register multiple PtoPI Interface instances for one source PI Data Archive with APS:

Method 1: The PtoPI_APS Connector supports a user-defined WHERE clause that selects the source PI Data Archive points which are candidate available points (see [Source Point Filter](#)). APS for each interface instance can be configured to “own” a specific subset of points from the source PI Data Archive so that there is no overlap in point ownership of the multiple interface instances. However, when devising the WHERE clauses, careful consideration must be used to ensure that the combined WHERE clauses for all interfaces catch all new points on the source PI Data Archive and that there is no overlap.

Method 2: The second method designates *one* PtoPI Interface instance to manage available points and its own existing points. The other PtoPI Interface instances associated with the same source PI Data Archive are restricted to updating their respective existing points. The essential element in this method is usage of the hidden points feature to configure the instance that manages available points to exclude the existing points of the other instances. The main disadvantage of this method is the difficulty of understanding the subtleties of maintaining the hidden points for the instance that manages available points. No WHERE clause is used with this method. Configure APS for all PtoPI Interface instances *except* the one designated to manage available points to update existing points and *not* to search for new (available) points. Configure APS for the PtoPI Interface instance designated to manage available points to update existing points and to search for new points. All new points added to the source PI Data Archive are automatically added to this instance of the PtoPI Interface. These points can be moved to other instances of the PtoPI Interface as needed.

Example 1: PtoPI Interface with Multiple Instances Using Method 1

Devise a plan for grouping all possible desired source PI Data Archive points into the PtoPI Interface instances. The goal is to configure APS for each instance of the PtoPI Interface so that each instance owns a specific subset of points possible on the source PI Data Archive and so that all points possible on the source PI Data Archive belong to the subset for one and only one interface instance. For example, source PI Data Archive points can be grouped by PointSource or by some pattern in the Tag name.

Example: If the plan is to bring all points from the source PI Data Archive to the target PI Data Archive that use specific PointSource values, like A, M, N, O, R, and L, and there are three instances of PtoPI, the points can be distributed in the following manner:

PtoPI Instance	WHERE clause
PtoPI1	pointsource = 'A*' or pointsource = 'M*'
PtoPI2	pointsource = 'N*' or pointsource = 'O*'
PtoPI3	pointsource = 'R*' or pointsource = 'L*'

Each instance would then update existing points and find new available points. After devising the WHERE clauses for assigning source PI Data Archive points to an instance of an interface, do the following two steps for each interface instance:

1. Perform this step for each PtoPI Interface. Run the APS Configuration Utility. Select one of the registered interfaces from the **Interface:** list. Click **Connector-specific...** on the **Settings** menu. In the **Filter for Available Source PI Server Points** area, select the **Retrieve only Available Points from the Source PI Server that satisfy the following WHERE clause** and then enter the WHERE clause. Use the **Test Expression** button to

confirm that the WHERE clause is a valid expression or synchronization will fail:

Filter for Available Points from Source PI Server strider

☒ Retrieve only Available Points from the Source PI Server that satisfy the following 'WHERE' clause:
(Default is to retrieve all Source PI Server points.)

PointSource = 'L' OR PointSource = 'R'

Retrieval Method for Source PI Server Points

☒ GetPoints ☐ GetPointsSQL

BA:ACTIVE.1
BA:PHASE.1
CDEP158
CDM158
CDT158
SINUSOID
SINUSOIDU

Test Expression

GetPoints found: 7
Status: Successful

2. Click **Rules...** on the **Settings** menu to open the **Rules for interface** dialog box. Select **Create automatically** in the **Points Not in PI** area and select **Edit automatically** in the **PI Point Edits** area:

Rules for PItoPI1 -> sandbox

These rules determine what the Sync Engine does with new points it finds, point changes it finds, and points that no longer exist on the data source.

Points Not in PI

☒ Create automatically ☐ Store in Available Points database (and optional log) ☐ Skip search for new points

PI Point Edits

☒ Edit automatically ☐ Store in PI Point Edits database (and optional log) ☐ Skip search for edits

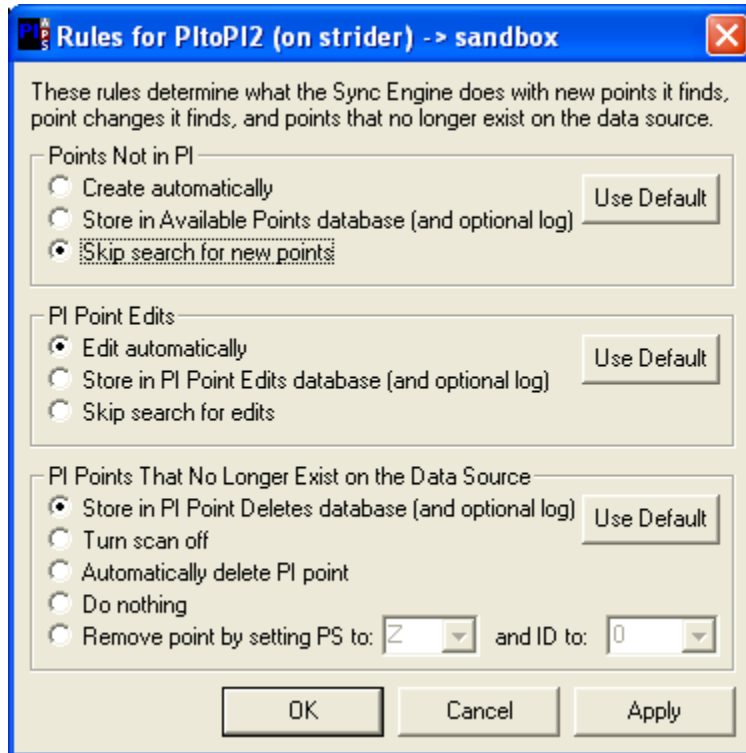
PI Points That No Longer Exist on the Data Source

☒ Store in PI Point Deletes database (and optional log) ☐ Turn scan off ☐ Automatically delete PI point ☐ Do nothing ☐ Remove point by setting PS to: and ID to:

This tells APS to automatically create new points and synchronize existing points.

Example 2: PItoPI Interface with Multiple Instances Using Method 2

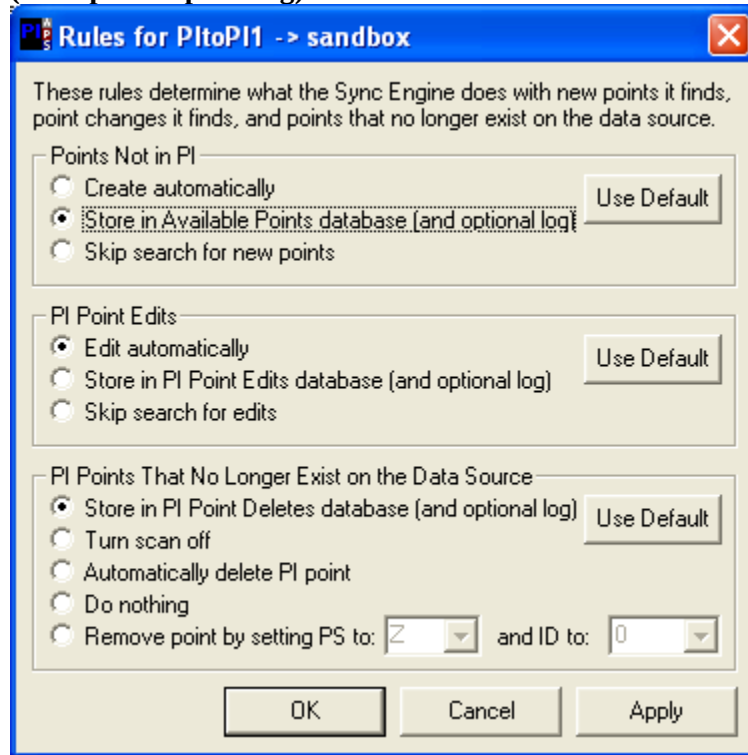
1. Perform this step for each registered PItoPI Interface instance *except* the instance designated to manage available points. Run the APS Configuration Utility. Select one of the PItoPI Interface instances from the **Interface:** list. Click **Rules...** on the **Settings** menu. On the **Rules for interface** dialog box, choose **Skip search for new points** in the **Points Not in PI** area:



This turns off the search for new points for each interface.

2. Perform this step and the following steps *only* for the PItoPI Interface instance *designated to manage available points*. Run the APS Configuration Utility. Select the designated PItoPI Interface instance from the **Interface:** list. Click **Rules...** on the **Settings** menu. On the **Rules for interface** dialog box, choose **Store in Available Points database**

(and optional point log).



3. Run a synchronization scan for the PItoPI Interface instance designated to manage available points.
4. In the APS Configuration Utility main window, select **Available points** and then click **Display Now**. The source PI Data Archive points are displayed. *Notice that the existing points that belong to the other PItoPI Interface instances appear in the list of available points.*
5. Review the available points and determine which available points belong to other PItoPI Interface instances. If *all* available points belong to other interface instances, right-click in the grid area, and then click **Hide All Points** on the shortcut menu.

If all available points do not belong to other interface instances, select the available points that belong to other interface instances. The grid area supports multiple selection. Right-click on a selected available point, and then click **Hide Points** to hide the selected points. This may be a tedious process and can be done in stages. As available points are hidden, they are removed from the available points list.
6. (Optional) After hiding all available points that are actually existing points for other interface instances, the **Points Not in PI** rule can be changed to **Create**

automatically:

Rules for PItoPI1 -> sandbox

These rules determine what the Sync Engine does with new points it finds, point changes it finds, and points that no longer exist on the data source.

Points Not in PI

- ☒ Create automatically Use Default
- ☐ Store in Available Points database (and optional log)
- ☐ Skip search for new points

PI Point Edits

- ☒ Edit automatically Use Default
- ☐ Store in PI Point Edits database (and optional log)
- ☐ Skip search for edits

PI Points That No Longer Exist on the Data Source

- ☒ Store in PI Point Deletes database (and optional log) Use Default
- ☐ Turn scan off
- ☐ Automatically delete PI point
- ☐ Do nothing
- ☐ Remove point by setting PS to: and ID to:

OK Cancel Apply

Decommissioned Source PI Data Archive Points

When a PI point is decommissioned, the history associated with the PI point frequently has long-term value. To preserve the history, a decommissioned PI point is not actually deleted from the PI Data Archive. Instead, a common practice is to logically delete the decommissioned point by changing attributes in some manner that stops collection of data. For example, the Scan attribute can be changed to off. Another common method for logically deleting a point is to change either the PointSource attribute or interface ID attribute to unused values, which disassociates the decommissioned point from any interface.

The PItoPI_APS Connector classifies an existing interface point as a candidate for deletion if and only if the mapping to a source PI Data Archive point fails. Since logically deleting a source PI Data Archive point does not break the mapping from an existing interface point, the PItoPI_APS Connector does not classify an existing interface point as a candidate for deletion when its source PI Data Archive point is logically deleted. In effect, the PItoPI_APS Connector cannot recognize logically deleted source PI Data Archive points and, therefore, the APS Synchronization Engine does not apply the rule for deleted points.

The recommended approach for logically deleting source PI Data Archive points is to change the Scan attribute to off. The reason for this recommendation is that the Scan attribute can be synchronized from a logically deleted source PI Data Archive point as a normal edit to an existing interface point. Obviously, the per-point synchronization settings must enable synchronization of the Scan attribute. When APS edits the Scan attribute to off for an existing interface point to synchronize with a logically deleted source PI Data Archive point, the PItoPI Interface stops scanning, which logically deletes the existing interface point on the target PI Data Archive.

Observe that changing the Scan attribute to off stops the PItoPI Interface from scanning but APS continues to synchronize the existing interface point with the source PI Data Archive point. If the source PI Data Archive point is recommissioned at a later time, APS will edit the Scan attribute to on, which causes the PItoPI Interface to resume scanning.

As long as the Scan attribute is changed to off for logically deleted source PI Data Archive points, other source PI Data Archive point attributes can also be changed when a source PI Data Archive point is logically deleted.

Changing PItoPI Interface /tn, /tnex, or /ptid Parameters

The PItoPI Interface /tn, /tnex, and /ptid parameters determine which interface point key attributes the PItoPI Interface and the PItoPI_APS Connector use to find either a source point tag name or ID, as explained in the [How PItoPI and PItoPI_APS Find Source Points](#) section. Adding or removing any of these parameters on the interface command line changes the key attributes that the PItoPI Interface and PItoPI_APS Connector use to map to source PI Data Archive points.



Caution: Before changing these parameters, you must ensure that all existing interface points will map correctly after the change. Otherwise, the interface may fail to load points and the PItoPI_APS Connector may mark points as deleted.

Before changing the interface parameters to ignore a key attribute, confirm that every existing point maps to the same source point by another key attribute. For example, assume that you are preparing to add the /tnex parameter to an interface instance, which causes the interface instance and the PItoPI_APS Connector to ignore InstrumentTag and UserInt1 as key attributes. Without the /tnex parameter, any interface point with a non-blank InstrumentTag value maps to the source point by tag name in InstrumentTag. For the existing point to map to the same source point after the /tnex parameter is added, the source point tag name must be in another key attribute (ExDesc or Tag). In this example, you must review every existing point with a non-blank InstrumentTag value and verify that ExDesc or Tag contains the same source point tag name. The PI Tag Configurator add-in for Excel is one tool that can be used to review and edit existing points before changing interface parameters.

Before changing the interface parameters to begin using a key attribute, confirm that every existing point has an attribute value that either does not map to the source point or maps to the same source point as the currently active key attributes. For example, assume that you are preparing to remove the /tnex parameter from an interface instance, which causes the interface instance to use InstrumentTag and UserInt1 as key attributes. For an existing interface point to map to the same source point after the /tnex parameter is removed, the InstrumentTag value must either be empty or contains the source point tag name. In this example, you must review every existing point and verify that the InstrumentTag value is either blank or contains the correct source point tag name. Alternatively, you can use a tool (like the PI Tag Configurator add-in for Excel) to change the InstrumentTag value for all existing points to blank. Another alternative is the change the InstrumentTag value for all existing points to the appropriate source point tag name, which the PItoPI_APS Connector can assist by enabling the option to store source point tag name in InstrumentTag and synchronizing *before* changing the interface parameters.

APS may assist in preparation for changing the interface parameters that determine the key attributes. Specifically when changing from ignoring a key attribute to using the key attribute, the PItoPI_APS Connector can be configured to store source point tag name or ID in the key attribute. Assuming that the per-point synchronization settings allow the attribute to be edited and the rule for point edits allows the APS Synchronization Engine to automatically edit, a synchronization scan will edit the about-to-be-enabled key attribute to a value that maps to the source point.

Appendix A. Error and Informational Messages

This appendix discusses several specific errors that can be encountered with the PItoPI_APS Connector and general troubleshooting methods for resolving other problems.

Installation Problems

Symptoms of installation problems are usually noticed when the APS Configuration Utility registers the first PItoPI Interface instance. Installation problems can also occur after the PItoPI_APS Connector is upgraded; the symptoms of post-upgrade installation problems are messages in various log files (see [Log Files](#) below for information on the log files).

The following is a list of symptoms and error messages, their explanations, and steps for resolving the circumstances that caused them.

Message	PItoPI_APS failed to register HRESULT = -2147024769
Cause	This message from the Windows Installer indicates that one or more prerequisite DLLs are missing.
Resolution	A common reason for this problem is that the APS Connector needs a newer prerequisite kit than is installed on the APS computer. Install a current prerequisite kit. This problem can also occur if required API libraries for the data source are not installed. Ensure that prerequisite software from the data source vendor is installed.

Symptom	PItoPI_APS does not appear as an installed connector in the APS Configuration Utility
Cause	The APS Connector must appear as "PItoPI_APS" in the Installed PI APS Connectors dialog box (Installed Connectors... on the Tools menu) and in the Select APS Connector: list in the Configure Interface or COM Connector for PI APS dialog box (Register New... on the Interface menu). If "PItoPI_APS" does not appear in these dialog boxes, PItoPI_APS.dll is not registered.
Resolution	To register PItoPI_APS.dll, open a command window (see Note 1) and enter the following commands: cd %PIHOME%\APS\Connectors\PItoPI_APS regsvr32 PItoPI_APS.dll

Message	Error 31023: The Connector-specific configuration tool PItoPI_APS_Config.PItoPI_APS_Config1 cannot be loaded: Invalid or unknown Class
Cause	If this message appears when a PItoPI Interface is selected and Connector-specific... is clicked on the Settings menu, PItoPI_APS_Config.dll is not registered.

Error and Informational Messages

Resolution	To register PItoPI_APS_Config.dll, open a command window (see Note 1) and enter the following commands: cd %PIHOME%\APS\Connectors\PItoPI_APS regsvr32 PItoPI_APS_Config.dll
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Message	PIAPSEngine.exe>PI-APS> Error from CoCreateInstanceEx. -2147221164: A specified class is not registered in the registration database. from call ptrAPSC.CreateInstance(PItoPI_APS.PIAPSCConnector) (for strider,Sandbox%PItoPI2) in LoadAPSRegisteredConnectors - will log every 100 errors PIAPSEngine.exe>PI-APS> Error from CoCreateInstanceEx. -2146233054: from call ptrAPSC.CreateInstance(PItoPI_APS.PIAPSCConnector) (for strider,Sandbox%PItoPI2) in LoadAPSRegisteredConnectors - will log every 100 errors PIAPSEngine.exe>PI-APS> Error from CoCreateInstanceEx. -2147024894: from call ptrAPSC.CreateInstance(PItoPI_APS.PIAPSCConnector) (for strider,Sandbox%PItoPI2) in LoadAPSRegisteredConnectors - will log every 100 errors
Cause	Other error numbers are also possible. The significant aspect of these messages is that the error was returned from a call to CreateInstance(PItoPI_APS.PIAPSCConnector). These messages indicate that PItoPI_APS.dll is not registered.
Resolution	To register PItoPI_APS.dll, open a command window (see Note 1) and enter the following commands: cd %PIHOME%\APS\Connectors\PItoPI_APS regsvr32 PItoPI_APS.dll

Message	The module "PItoPI_APS.dll" was loaded but the call to DllRegisterServer failed with error code 0x80004005.
Cause	Regsvr32 does not have privileges to change the registry.
Resolution	See Note 1.

Note 1: Log in as an Administrator to run regsvr32. On Windows Vista or later that include User Account Control (UAC), right-click the cmd utility and select **Run as administrator** on the shortcut menu to open a command window with Administrator privileges.

Upgrade Problems

Following an upgrade, if the logs indicate that the old version of the PItoPI_APS Connector is still being used, the APS Synchronization Engine needs to be stopped and restarted. When the APS Synchronization Engine starts, it loads copies of the APS Connectors for all registered interface instances into its virtual memory. After an APS Connector is loaded, the APS Synchronization Engine uses the copy in its virtual memory until the APS Synchronization Engine service is stopped. Thus, if an APS Connector is upgraded while the APS Synchronization Engine is running, the new version will not be used until the APS Synchronization Engine is restarted. Follow the steps in the [Upgrading Checklist](#) section to restart the APS Synchronization Engine.

Operational Problems

Log Files

When the APS Synchronization Engine performs a synchronization scan for an interface, it creates a log file for that synchronization scan. If errors occur during the synchronization scan, the APS Synchronization Engine writes error details to the `pipc.log` file and usually also writes an error message into the log file for the synchronization scan. The log files created for the synchronization scans must be routinely monitored. If indications of errors are found, additional information may be available in `pipc.log`. The APS Configuration Utility provides simple access to these log files. On the **Tools** menu, click **Log Files...** to open the **Synchronization & Point Logs** dialog box.

In normal operation, the PItoPI_APS Connector rarely logs messages. When it does, the message is written to the log file for the synchronization scan. When the PItoPI_APS Connector encounters an error that it cannot handle, it returns an error code and description string to the APS Synchronization Engine, which the APS Synchronization Engine usually writes to both the `pipc.log` file and the log file for the synchronization scan.

The APS Configuration Utility typically reports errors and unusual conditions in the status bar on the main window or a message area on a dialog box. The APS Configuration Utility also writes a message to the PI SDK log (not `pipc.log`) for error conditions.

Operational Errors

For operational errors that are common to all APS Connectors, see *Appendix A: Troubleshooting* in the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide. The following list contains known error messages that are specific to the PItoPI_APS Connector, their explanations, and steps for resolving the circumstances that caused those messages to be displayed.

Message	Error -2147200248: Illegal configuration settings for interface or PItoPI_APS (ptrAPSC->GetUpdatedAttributes in PerformASynchronization)
Cause	This error number indicates that the PItoPI_APS Connector cannot proceed because of either errors in its configuration or incompatibility with certain interface parameters. This message is preceded by a message that provides more specific information.
Resolution	Review the messages that precede this message in the log files.

Message	Illegal interface configuration: both <code>/ptid</code> and <code>/sec_src</code> parameters are present.
Cause	The PItoPI Interface and PItoPI_APS Connector do not support identification of source points by point ID (<code>/ptid</code>) when source PI Data Archive-level failover (<code>/sec_src=</code>) is enabled.
Resolution	The PItoPI Interface is also nonfunctional when the interface parameters contain both <code>/ptid</code> and <code>/sec_src=</code> . Correct the PItoPI Interface configuration. The Connector-specific settings for the PItoPI_APS Connector must be reviewed after changing the PItoPI Interface configuration.

Error and Informational Messages

Message	This version of PIttoPI_APS includes PtSecurity and DataSecurity in its Known Attributes, which requires Sync Engine version 1.2.5.0 or later. Error -2147200247: PIttoPI_APS requires Sync Engine 1.2.5.0 or later. (ptrAPSC->GetUpdatedAttributes in PerformASynchronization)
Cause	This error number indicates that the PIttoPI_APS Connector cannot proceed because the known attributes include the new security model attributes that were introduced in PI Data Archive version 3.4.380. APS Connectors that support the new PtSecurity or DataSecurity attributes cannot be used with APS Synchronization Engine versions earlier than 1.2.5.0.
Resolution	Upgrade APS to version 1.2.5.0 or later to support PIttoPI_APS Connector versions 1.3.2.0 and later.

Message	In GetAvailablePoints, GetPoints returned -2147220325. No matching points found on the Source PI Server strider. WHERE clause: PointSource = 'UnusedPointSource'.
Cause	No points in the source PI Data Archive satisfy the conditions in the WHERE clause. Therefore, there are no candidate available points. Although this message is not necessarily an error, this message usually indicates that the WHERE clause is not what you intended.
Resolution	Verify the WHERE clause and revise as needed.

Message	In GetAvailablePoints, GetPointsSQL returned -2147220325. No matching points found on the Source PI Server strider. SQL WHERE clause: PointSource = 'UnusedPointSource'.
Cause	No points in the source PI Data Archive satisfy the conditions in the WHERE clause. Therefore, there are no candidate available points. Although this message is not necessarily an error, this message usually indicates that the WHERE clause is not what you intended.
Resolution	Verify the WHERE clause and revise as needed.

Message	Cannot create a point for a source tag longer than 80 characters because the interface ignores source PointID in UserInt1. Source tag: BA_name_longer_than_80_characters01234567890123456789012345678901234567890123456789
Cause	The PIttoPI_APS Connector found an available source point with a tag name longer than 80 characters. The only way to configure an interface point for a source point that has a tag name longer than PI API supports is by source point ID in UserInt1. However, this is not possible because the interface parameters include <i>/tn</i> or <i>/tnex</i> , which cause the interface to ignore UserInt1 as a key attribute.
Resolution	Interface points for source points with tag names longer than 80 characters are not possible when the interface parameters include <i>/tn</i> or <i>/tnex</i> .

Message	Cannot create a point for a source tag longer than 80 characters because the PItoPI_APS option to store source PointID in UserInt1 is cleared. Source tag: BA_name_longer_than_80_characters01234567890123456789012345678901234567890123456789
Cause	The PItoPI_APS Connector found an available source point with a tag name longer than 80 characters. The only way to configure an interface point for a source point that has a tag name longer than PI API supports is by source point ID in UserInt1. However, this is not possible because the PItoPI_APS option to store source point ID in UserInt1 is not selected.
Resolution	Interface points for source points with tag names longer than 80 characters are not possible when the PItoPI_APS option to store source point ID in UserInt1 is not selected.

Message	Cannot create a point for a source tag longer than 80 characters because source PI failover causes the interface to reject source PointID in UserInt1. Source tag: BA_name_longer_than_80_characters01234567890123456789012345678901234567890123456789
Cause	The PItoPI_APS Connector found an available source point with a tag name longer than 80 characters. The only way to configure an interface point for a source point that has a tag name longer than PI API supports is by source point ID in UserInt1. However, this is not possible because the interface is configured for source PI Data Archive-level failover.
Resolution	Interface points for source points with tag names longer than 80 characters are not possible when the interface is configured for source PI Data Archive-level failover.

Appendix B. Terminology

To understand this user guide, you must be familiar with the terminology used in this appendix. Specifically, you must have read the *PI AutoPointSync for Interfaces and PI COM Connectors* user guide that explains important PI AutoPointSync concepts and provides full definitions of PI AutoPointSync terminology.

PI AutoPointSync Terms

APS

An acronym for PI AutoPointSync.

APS Configuration Utility

The interactive application that registers instances of interfaces for automatic point synchronization and configures APS options.

APS Connector

An interface-specific module that communicates with the same data source as the interface to obtain current tag attribute values. Generally, an APS Connector is designed for a particular interface and its implementation is based on a specific programming interface for the data source. During each synchronization scan, the APS Synchronization Engine dynamically loads the APS Connector for the interface instance and calls it to retrieve information for updating the PI point database.

APS Connector-specific Control

A module that provides the user interface for configuring the options of a specific APS Connector. Most APS Connectors have companion APS Connector-specific Controls. The APS Configuration Utility dynamically loads the APS Connector-specific Control that is identified by the APS Connector for the selected interface instance.

APS Synchronization Engine

The core service of the AutoPointSync product that schedules and performs synchronization scans for registered interface instances. During a synchronization scan, it obtains current tags and their attributes from a data source by dynamically loading and calling “plug-ins” known as APS Connectors. The APS Synchronization Engine creates, edits, or deletes PI points as necessary to agree with current tag definitions in the data source.

APS Synchronization Trigger Service

A service that monitors for events that are configured to trigger a synchronization scan for a registered interface instance.

Available Points

PI points that can be created for data source tags that are not associated with a PI point. Available points do not exist in the PI point database.

Existing Points or Existing PI Points

PI points that already exist and are assigned to an interface instance that is registered with APS.

Hidden Points

Former available points that someone marked for APS to ignore. Hidden points do not exist in the PI point database. They are excluded from the display of available points and prevent creation of PI points for the associated data source tags.

Key Attributes

The PI point attributes that are required by the interface and APS to identify and access the data source tag for the PI point.

Known Attributes

The attributes for which an APS Connector can provide values when a PI point is created.

Master Synchronization Setting

A per-point configuration option that controls whether a PI point is synchronized by APS. Synchronization is enabled or disabled individually for each existing PI point.

PI Point Deletes

Existing PI points that no longer have a valid data source tag.

PI Point Edits

Attributes of existing PI points that are not in agreement with the corresponding data source tag attributes.

Syncable Attributes

Attributes for which an APS Connector can provide values to update an existing PI point.

Synchronization or Synchronization Scan

The process that compares PI points for an interface with data source tags and either changes the PI Data Archive to resolve any differences or logs the differences.

General Terms

Attribute

A parameter that describes a PI point. Each PI point has an associated list of attributes. Some attributes are simply descriptive. Other attributes are configuration parameters for the PI Data Archive, the interface instance that transfers data between the PI point and data source, or both.

COM Connector

A COM object that allows the PI Data Archive to access data from a foreign data historian and make it available to any PI client application in a seamless fashion. Some currently available COM Connectors include those for data historians from AspenTech and Honeywell as well as one for any data source with an OLEDB provider. COM Connectors are only available on Windows platforms.

DCS

An acronym for Distributed Control System. DCS was used in earlier versions of APS documentation as a generic term for any data source that provides data to a PI Data Archive, including measurement and control systems or other historian systems, even other PI Data Archives.

ICU

The PI Interface Configuration Utility. The ICU is the primary application that you use to configure PI interface programs. You must install the ICU on the same computer on which an interface runs. A single copy of the ICU manages all of the interfaces on a particular computer.

You can configure an interface by editing a startup command file. However, APS cannot access interface parameters from a startup command file. Therefore, configuring an interface instance with the ICU is a prerequisite to registering the interface instance with APS.

Interface

A software program that collects data from some type of data source and sends the data to a PI Data Archive. Some interfaces also have the ability to read data from a PI Data Archive and write back to the data source.

Interface Node

A computer on which

- the PI API and/or PI SDK are installed, and
- PI Data Archive programs are not installed.

PI API

A library of functions that PI client applications call to communicate and exchange data with the PI Data Archive. All PI interfaces use the PI API.

PI Data Archive Collective

Two or more replicated PI Data Archives that collect data concurrently. Collectives are part of the High Availability environment. When the primary PI Data Archive in a collective

becomes unavailable, a secondary collective member seamlessly continues to collect and provide data access to your PI clients.

PI Data Archive Node

A computer on which PI Data Archive programs are installed. The PI Data Archive runs on the PI Data Archive node. In earlier documentation, PI Data Archive was referred to as the PI Server (see [PI Server Node](#)).

PIHOME

The directory that is the common location for 32-bit PI client applications. OSIsoft installation kits create a system environment variable named `PIHOME` that is set to the path of the top-level directory for 32-bit PI clients.

A typical `PIHOME` on a 32-bit operating system is `C:\Program Files\PIPC`.

A typical `PIHOME` on a 64-bit operating system is `C:\Program Files (x86)\PIPC`.

PI interfaces reside in a subdirectory of the `Interfaces` directory under `PIHOME`. Most APS files are in the `APS` directory under `PIHOME`.

pipc.log

The file to which OSIsoft applications write informational and error messages. While a PI interface runs, it writes to the `pipc.log` file. The ICU and APS Configuration Utility provide easy access to the `pipc.log`. The `pipc.log` file is in the `dat` subdirectory of `PIHOME`.

PI SDK

A library of functions that client applications call to communicate and exchange data with the PI Data Archive. Some PI interfaces, in addition to using the PI API, require the use of the PI SDK. APS uses PI SDK to communicate with the PI Data Archive.

PI Server Node

A computer on which PI Data Archive programs are installed. In earlier documentation, the term “PI Server” was used as a nickname for the PI Data Archive and a PI Server node was a computer on which PI Data Archive programs were installed. While the PI Data Archive remains a core server of the PI Server product, the product name “PI Server” now refers to much more than the PI Data Archive. OSIsoft documentation, including this user guide, is changing to use “PI Server” in this broader sense and “PI Data Archive” to refer to the historian core (see [PI Data Archive Node](#)).

PI SMT

PI System Management Tools. PI SMT is the program that you use for configuring PI Data Archives. A single copy of PI SMT manages multiple PI Data Archives. PI SMT runs on either a PI Data Archive node or a PI Interface node.

Point

The basic building block for controlling data flow to and from the PI Data Archive. For a given timestamp, a PI point holds a single value.

A PI point does not necessarily correspond to a "point" on the data source. For example, a single "point" on the data source can consist of a set point, a process value, an alarm limit, and a discrete value. These four pieces of information require four separate PI points.

In APS documentation, *point* means a PI point and *tag* designates a “point” on the data source.

Service

A Windows program that runs without user interaction. A service has the ability to start up when the computer itself starts up. If started manually, it continues to run after you log off from Windows.

Tag (Input Tag and Output Tag)

The name of the PI point. There is a one-to-one correspondence between the name of a point and the point itself. Because of this relationship, PI System documentation uses the terms "tag" and "point" interchangeably.

In APS documentation, *point* means a PI point and *tag* designates a “point” on the data source.

Interfaces read values from a data source and write these values to an Input Tag. Interfaces use an Output Tag to write a value to the data source.

UniInt (Universal Interface)

A framework for interfaces to the PI Data Archive. UniInt provides common features and generic functions required by most interfaces. Most interfaces developed by OSIsoft are based on the UniInt framework, which results in a consistent set of features in the OSIsoft interfaces. For example, the UniInt framework supports special points that can be used to monitor the performance and health of an interface. The UniInt framework also supports a failover mechanism for interface redundancy. The *UniInt Interface User Manual* provides complete information about the features provided by UniInt.

Appendix C. Technical Support and Resources

For technical assistance, contact OSIsoft Technical Support at +1 510-297-5828 or the [OSIsoft Customer Portal Contact Us page \(https://customers.osisoft.com/s/contactus\)](https://customers.osisoft.com/s/contactus). The Contact Us page offers additional contact options for customers outside of the United States.

When you contact OSIsoft Technical Support, be prepared to provide this information:

- Product name, version, and build numbers
- Details about your computer platform (CPU type, operating system, and version number)
- Time that the difficulty started
- Log files at that time
- Details of any environment changes prior to the start of the issue
- Summary of the issue, including any relevant log files during the time the issue occurred

To ask questions of others who use OSIsoft software, join the OSIsoft user community, [PI Square](#). Members of the community can request advice and share ideas about the PI System. The [PI Developers Club](#) space within PI Square offers resources to help you with the programming and integration of OSIsoft products.